Beyond Dice: Evaluating U-Net and nnU-Net for Mitochondria Segmentation in Electron Microscopy Using Biologically Motivated Metrics

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Abstract

Automated segmentation of mitochondria in electron microscopy images is increasingly used to support biological analyses that depend on individual organelle counts, sizes, and spatial organisation. Standard evaluation practice relies on the Dice similarity coefficient and related pixel-level overlap metrics. These quantify foreground overlap but are insensitive to object-level errors such as merges, splits, and incorrect counts. The dissertation tests whether standard overlap metrics are sufficient for evaluating model quality in mitochondria segmentation. It compares a custom U-Net with the self-configuring nnU-Net framework on the CEM-MitoLab dataset. Both models are evaluated under a shared biological source-level split and a common pipeline spanning pixel-level, boundary, calibration, and object-level metrics.

nnU-Net outperforms the custom U-Net across all metric families (foreground Dice 0.624 vs. 0.578; Panoptic Quality 0.544 vs. 0.459). The object-level gap is larger than the pixel-level gap, and is clearest in dense images where foreground Dice draws the models close to equivalence while Panoptic Quality and split rate distinguish the models. Among high-Dice predictions, 42.3% of U-Net slices and 38.1% of nnU-Net slices still showed object-level errors. U-Net failures are mainly fragmentation, while nnU-Net failures are mainly merging. Standard overlap metrics cannot distinguish between these failure modes, making object-level evaluation necessary to reliably assess model quality for biological use.

Research Ethics Approval

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Declaration

I declare that this thesis was composed by myself, that the work contained herein is my own except where explicitly stated otherwise in the text, and that this work has not been submitted for any other degree or professional qualification except as specified.

(*Mira Kawar*)

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Chapter 1

Introduction

Mitochondria are essential organelles that support cellular energy production, cell survival, and metabolic regulation. Their morphology, including size, shape, and overall organisation, is closely linked to cellular function (Suomalainen and Nunnari, 2024). Disruption of mitochondrial structure is implicated in a range of severe diseases, including Parkinson's disease, Alzheimer's disease, diabetes, and cancer (Suomalainen and Nunnari, 2024; Tilokani et al., 2018). Accurate, large-scale quantification of mitochondrial morphology in biological tissue therefore has direct relevance to both basic cell biology and biomedical research.

Electron microscopy (EM) is widely used to observe mitochondrial ultrastructure at nanometre scale, revealing internal organisation and structural details that light microscopy cannot capture. However, EM datasets are large, heterogeneous, and labour-intensive to annotate manually, which has driven the development of automated deep learning methods for mitochondria segmentation (Aswath et al., 2023). These methods have advanced rapidly, from early patch-based convolutional networks to the encoder-decoder U-Net architecture (Ronneberger et al., 2015) and, more recently, the self-configuring nnU-Net framework (Isensee et al., 2021), which adjusts its setup automatically based on dataset properties.

1.1 Problem, Aim, and Research Question

Despite this progress, a central problem in comparing and deploying segmentation models remains unresolved. Standard pixel-overlap metrics such as Dice and Intersection over Union (IoU) measure pixel-level agreement between prediction and ground truth, but they are largely insensitive to the spatial organisation of individual objects. A model can achieve a high overlap score while still fragmenting a single mitochondrion into multiple predicted components, merging adjacent mitochondria into one, or misestimating the total number of mitochondria in a field of view. Such errors directly distort biological analyses that rely on organelle count, size distribution, and spatial organisation (Suomalainen and Nunnari, 2024; Wei et al., 2020; Conrad and Narayan, 2023). Existing comparisons in the literature are also rarely conducted under compatible experimental setups: models differ in supervision target, dataset, dimensionality, and

evaluation protocol, which makes it difficult to separate differences in model behaviour from differences in experimental design (Wei et al., 2020; Conrad and Narayan, 2023; Xie et al., 2025; Chen et al., 2025).

This dissertation investigates whether standard pixel-level metrics are sufficient for comparing segmentation models in a biology-facing setting. The study uses a controlled comparison between a custom U-Net and nnU-Net on the CEM-MitoLab electron microscopy dataset (Narayan, 2022), and is organised around three aims. First, it tests whether standard pixel-level metrics can understate biologically meaningful differences between models. Second, it examines whether pixel-level metrics adequately reflect distinct biologically relevant error patterns such as merges and splits. Third, it tests whether a model can achieve a strong pixel-level score while still making errors that would matter in biological analysis.

This leads to the following research question:

> *To what extent are standard pixel-level metrics, particularly the Dice similarity coefficient, sufficient for comparing a custom U-Net and nnU-Net on mitochondria segmentation in electron microscopy, and which biologically relevant failure modes do they fail to capture?*

1.2 Overview of Findings

nnU-Net outperforms the custom U-Net overall, with a modest but consistent pixel-level advantage. The object-level separation is larger, and clearest in dense images, where foreground Dice narrows while Panoptic Quality and split rate continue to distinguish the models. Among high-Dice predictions, 42.3% of U-Net slices and 38.1% of nnU-Net slices still show object-level errors, with the nature of those errors depending on the model. Standard overlap metrics do not capture this distinction.

1.3 Contributions

This dissertation makes the following contributions:

- A **standardised comparison pipeline** in which both a custom U-Net and nnU-Net are trained on the same biological source-level data split and evaluated with the same post-processing code, enabling a fairer direct comparison.
- A **biologically motivated evaluation framework** combining pixel-level metrics (Dice, IoU, precision, recall, HD95, boundary F1, Expected Calibration Error) with object-level biological metrics (object count error, merge rate, split rate, Panoptic Quality) derived from the dataset's original instance annotations.
- A **corner-case analysis** that identifies slices with high Dice (> 0.85) but elevated object-level error, showing that strong overlap can still conceal biologically important failures.

- A **stratified analysis** showing how model behaviour changes across different image conditions, particularly mitochondrial foreground density.

1.4 Dissertation Structure

The rest of the dissertation is organised as follows:

- Chapter 2 reviews the relevant background and literature on EM segmentation methods, datasets, and related approaches.
- Chapter 3 presents the evaluation metrics and framework used in the dissertation.
- Chapter 4 describes the methodology, including data preparation, model training, and the shared evaluation framework.
- Chapter 5 presents the results and analysis for the custom U-Net, nnU-Net, and the final comparison between them.
- Chapter 6 summarises the main findings and discusses limitations and future work.

Chapter 2

Background and Related Work

Electron microscopy (EM) is valuable for mitochondria segmentation because it provides the spatial detail needed to resolve organelle boundaries and internal structure (Aswath et al., 2023; Conrad and Narayan, 2023). That same detail also makes the task difficult. Mitochondria vary in size, shape, contrast, and local context, and they are often tightly packed or touching (Aswath et al., 2023; Wei et al., 2020). For that reason, segmentation errors are often biologically meaningful: they appear as merges, splits, or incorrect object counts (Wei et al., 2020; Franco-Barranco et al., 2022; Conrad and Narayan, 2023).

A framework that reports high pixel overlap is not automatically solving the biologically relevant problem (Wei et al., 2020; Conrad and Narayan, 2023). Foreground-background classification and reliable recovery of individual mitochondria as separate objects are related, but not equivalent. A model may recover most mitochondria pixels correctly while still distorting the number or shape of individual organelles. The central question is whether methods that score well on benchmarks also produce segmentations that are reliable for object-level biological analysis (Wei et al., 2020; Franco-Barranco et al., 2022; Conrad and Narayan, 2023).

2.1 Dataset Choice

Smaller datasets such as Lucchi and the re-annotated Lucchi++ benchmark have been useful for benchmarking. Their main strength is that they support clean, repeatable comparisons with relatively stable labels. Their main weakness is limited scope. Reported overlap scores on these datasets are already high, and they cover a comparatively narrow range of biological sources, imaging conditions, and mitochondrial appearances. Success on Lucchi-style data therefore provides only limited evidence of robustness across more diverse settings (Lucchi et al., 2012; Casser et al., 2020; Franco-Barranco et al., 2022; Conrad and Narayan, 2023).

MitoEM presents a more challenging benchmark for the specific failure modes considered in this dissertation. Its authors report around 40K instances with great diversity in shape and density, and show that existing methods often fail on complex shapes or

densely packed instances (Wei et al., 2020). That makes MitoEM a better diagnostic dataset than Lucchi for separation failures. At the same time, MitoEM is still derived from human and rat cortex volumes, whereas CEM-MitoLab was assembled to cover a broader range of tissues, organisms, imaging techniques, and resolutions (Wei et al., 2020; Conrad and Narayan, 2023). That narrower biological scope makes MitoEM less suitable than CEM-MitoLab for asking how general-purpose semantic pipelines behave across varied sources.

CEM-MitoLab is the most relevant dataset in this comparison because it was constructed as a deliberately heterogeneous mitochondria benchmark, combining legacy labels and crowdsourced annotations. The published dataset description reports 21,860 annotated 2D images (Conrad and Narayan, 2023). However, the downloaded dataset used in this dissertation contained 21,871 images, and all splits, training, and evaluation were performed on that downloaded version. These images contain 135,283 mitochondrial instances drawn from varied image sizes, resolutions, imaging techniques, cells, tissues, and organisms. That breadth is a strength because it makes the comparison less dependent on a single narrow imaging regime. The trade-off is that heterogeneity also introduces greater variation in label style, image quality, and local difficulty, so raw overlap scores are harder to interpret in isolation.

2.2 Segmentation Approaches in the Literature

The literature on mitochondria segmentation includes several model families (Aswath et al., 2023; Conrad and Narayan, 2023). A useful distinction is between semantic models, which assign pixels to foreground or background, and methods that explicitly target instance separation. Recent work also places greater emphasis on generalisation across datasets and imaging conditions (Xie et al., 2025; Huang et al., 2025).

2.2.1 U-Net as the Standard Semantic Baseline

U-Net, introduced by Ronneberger et al. (2015), is an encoder-decoder convolutional network designed for dense biomedical image segmentation. The contracting path extracts increasingly abstract features, while the expanding path upsamples those features back toward the input resolution. Skip connections link corresponding encoder and decoder stages so that fine spatial detail is preserved during reconstruction. These design choices helped make U-Net a standard baseline in biomedical segmentation (Ronneberger et al., 2015; Aswath et al., 2023).

For mitochondria segmentation, U-Net remains relevant because its variants still appear widely in mitochondria segmentation papers, including real-time 2D systems, boundary-aware 3D pipelines, and more recent automated EM segmentation workflows (Casser et al., 2020; Wei et al., 2020; Jang et al., 2025). One major strength of U-Net is that it is clear, well understood, and easy to adapt. At the same time, a reported “U-Net baseline” is often less standard than it appears. Patch size, normalisation, augmentation, loss weighting, and optimisation can all change performance substantially. Improvements over a U-Net baseline sometimes reflect differences in training setup, tuning, and implementation choices (Isensee et al., 2021; Franco-Barranco et al., 2022).

There is also a task-specific limitation. Standard U-Net predicts a semantic foreground mask. Separate mitochondria can be recovered afterwards with connected-component analysis or other post-processing, but the network itself does not preserve object identity. Merge and split errors therefore remain an expected failure mode even when pixel overlap is strong (Wei et al., 2020; Conrad and Narayan, 2023).

2.2.2 nnU-Net as an Automated Baseline

nnU-Net, introduced by Isensee et al. (2021), is a self-configuring segmentation framework. It is not a single fixed model. It starts from a dataset fingerprint, including image size, spacing, class balance, and intensity statistics, and uses a set of heuristic rules to choose the preprocessing pipeline, architecture configuration, patch size, training schedule, and inference procedure. Depending on the dataset, this may lead to a 2D U-Net, a 3D U-Net, or a cascade (Isensee et al., 2021; Çiçek et al., 2016).

The main strength of nnU-Net is standardisation. It reduces the number of manual design choices that shape performance and has become a widely used reference point in biomedical segmentation because it performs strongly across a large range of tasks (Isensee et al., 2021). It is therefore important in the current literature both as a practical tool and as evidence that a large share of segmentation performance comes from proper engineering decisions (Isensee et al., 2021).

nnU-Net also has limitations. It can be computationally heavy, especially in its full five-fold ensemble form, and it is still usually judged by benchmark overlap scores. High semantic segmentation accuracy under that standard does not necessarily mean that individual mitochondria are segmented in a way that preserves correct counts, boundaries, or separation between neighbouring objects (Wei et al., 2020; Conrad and Narayan, 2023).

2.2.3 Instance-Aware Methods and What They Show

A substantial part of the mitochondria segmentation literature moves beyond binary semantic masks and targets instance separation directly. This is usually done with pipelines that predict masks, boundaries, or auxiliary shape cues and then apply post-processing to recover individual instances (Wei et al., 2020; Conrad and Narayan, 2023). These methods matter here because they show that separating individual mitochondria is a different problem from recovering foreground pixels. In this chapter they serve mainly as motivation for object-level evaluation, not as direct baselines. Many such methods change the supervision target, task formulation, dimensional setting, or post-processing pipeline, so they would not support a controlled semantic-baseline comparison.

Wei et al. (2020) make this point well through the U3D-BC and U3D-BC+MW pipelines, which combine a boundary-aware 3D U-Net with marker-controlled watershed post-processing. These models are effective because they explicitly target touching mitochondria and do not treat all foreground as one class. This is relevant to the dissertation because it demonstrates that strong semantic overlap alone is insufficient to guarantee accurate recovery of individual objects.

Conrad and Narayan (2023) make a similar contribution through MitoNet, a generalist instance segmentation model built on Panoptic-DeepLab (Cheng et al., 2020) and trained on the diverse CEM-MitoLab and CEM1.5M resources (Conrad and Narayan, 2023). It pairs a generalist instance model with broad training data and challenging unseen benchmarks. More importantly, it shows what more extensive instance-labelled data and broader sampling can enable, but it is not a controlled comparison between semantic baselines.

Shape-aware and transformer-based methods push this further by encoding priors about mitochondrial geometry or long-range structure (Aswath et al., 2023; Chen et al., 2025). Their advantage is that they are designed for precisely the elongated, tightly packed, or fragmented structures that cause merge and split errors. This is a real improvement over plain semantic masking when the goal is accurate instance separation. They show that instance-aware performance can be improved, but they do not make it easy to isolate whether the gain comes from architecture, supervision, dimensionality, or evaluation protocol.

2.2.4 Recent Trends in the Literature

Recent work has moved in two directions that are relevant for interpreting the state of the field. One direction aims for broader generalisation. For example, Xie et al. (2025) show that transfer-learning pipelines can improve organelle segmentation across tissues, while Huang et al. (2025) emphasise a generalist deep-learning segmentation tool for volume electron microscopy. This line of work addresses a weakness in older benchmark culture: models that look strong on one curated dataset may perform much less reliably on data from different biological sources or imaging conditions. It is more relevant to questions of generalisation and practical usability than to the evaluation of biological reliability.

A second direction is increasing task sophistication. Papers such as Jang et al. (2025) and Chen et al. (2025) continue to improve segmentation quality through stronger architectures, improved context modelling, or more specialised post-processing. Jang et al. (2025) present an automated mitochondrial segmentation pipeline for complex transmission electron microscopy images, while Chen et al. (2025) introduce GCTransNet, a 3D transformer-based pipeline.

This point is reinforced by broader organelle-segmentation work such as Heinrich et al. (2021), which shows that large-scale EM segmentation can benefit from substantial annotation effort, specialised pipelines, and application-specific engineering. This literature demonstrates the level of performance achievable in well-resourced settings. However, it does not provide a close methodological template for the present comparison because it often combines multiple organelles, 3D context, and highly customised workflows. These papers mainly provide background context for the field. They are not strong direct baselines for a mitochondria-specific comparison. They show where the field is moving, but they do not provide a clean basis for judging whether two semantic mitochondria-segmentation pipelines differ in biologically meaningful ways.

Chapter 3

Evaluation Metrics and Framework

3.1 Why Overlap Metrics Persist

Dice and IoU dominate the segmentation literature because they are simple, interpretable, and widely reported across papers. Dice is especially popular in biomedical segmentation because it is less harsh than IoU under class imbalance and is often used directly as both a training objective and a reporting metric (Taha and Hanbury, 2015; Franco-Barranco et al., 2022; Isensee et al., 2021). Precision, recall, and boundary-sensitive measures such as HD95 provide additional detail about false positives, false negatives, and boundary error (Taha and Hanbury, 2015).

The Dice similarity coefficient for a predicted mask P and ground-truth mask G is defined as:

$$\text{Dice} = \frac{2|P \cap G|}{|P| + |G|}$$

and the Intersection over Union (IoU) is:

$$\text{IoU} = \frac{|P \cap G|}{|P \cup G|}$$

Both range from 0 to 1, but IoU is stricter because it penalises disagreement more heavily for the same prediction (Taha and Hanbury, 2015). When written in terms of true positives (TP), false positives (FP), and false negatives (FN), pixel-level precision and recall are:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}, \qquad \text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

which makes clear that they separate over-prediction from under-prediction in a way Dice alone does not (Taha and Hanbury, 2015).

These metrics are useful, but they all describe agreement at the pixel level. On their own, they do not show whether the model has identified the correct mitochondria as distinct objects (Wei et al., 2020; Conrad and Narayan, 2023).

Comparison is also complicated because Dice can be aggregated in several valid ways and can change with thresholding or post-processing choices. It may refer to foreground

Dice, mean Dice over classes, per-image averages, or dataset-level aggregates. Unless those choices are stated clearly, two papers can both report Dice without reporting exactly the same quantity (Taha and Hanbury, 2015).

Several pixel-level summaries are useful because each one answers a slightly different question. Foreground Dice is the main overlap measure because it reflects agreement on the mitochondria class directly. Background Dice is usually easier to satisfy, especially in sparse images, so it is useful mainly for checking whether performance is being inflated by the dominance of background pixels. Mean Dice averages the two classes, which can be convenient as a balanced summary but may still hide the fact that foreground quality is the harder part of the task.

3.2 Why Overlap Metrics Are Not Enough

Published segmentation studies already show this problem in practice. Franco-Barranco et al. (2022) show that overlap-based benchmark scores on established datasets can become very similar even when qualitative differences in the predictions remain visible. When that happens, small score differences are harder to interpret as meaningful practical improvement. Wei et al. (2020) make a related point from the instance-segmentation side: boundary-aware methods are needed because overlap scores alone do not show whether touching mitochondria have been separated correctly. These papers suggest that overlap metrics can flatten differences that matter for biological use.

For mitochondria segmentation, this matters because many biological analyses are instance-dependent. If a model merges neighbouring mitochondria, measured counts decrease and object sizes inflate. If it splits one mitochondrion into many parts, counts increase and morphology statistics become unreliable. These two errors can have opposite biological implications while still producing broadly similar overlap scores. Dice and IoU therefore answer only part of the question. They tell us how well foreground pixels match, but not whether the model has preserved the object structure that many biological analyses rely on (Wei et al., 2020; Conrad and Narayan, 2023).

The presented pixel-level evaluation also examines threshold sensitivity by repeating foreground Dice across several binarisation thresholds. This is useful because a model can appear strong at the default threshold while depending on unusually fragile probability calibration (Guo et al., 2017). In this work, threshold sensitivity is used as a practical robustness check. It is not intended as a standard literature metric. If a model's apparent quality changes sharply under small threshold adjustments, that makes its overlap score harder to trust in use. It helps distinguish models whose reported overlap is stable from those whose apparent quality is threshold-dependent.

3.3 Object-Level Metrics and What They Add

Object-level metrics address this gap by evaluating predicted instances directly. They go beyond foreground-pixel agreement alone. Merge rate shows how often distinct ground-truth mitochondria are fused into one prediction, while split rate shows how often one

ground-truth mitochondrion is broken into multiple predicted objects. Connectivity error is defined as the sum of merge and split events, divided by the number of ground-truth objects. Object count error measures the practical consequence of those failures for count-based analysis. Panoptic Quality (PQ) goes further by combining detection and segmentation quality into a stricter instance-level summary (Kirillov et al., 2019).

In its standard form,

$$\mathrm{PQ} = \underbrace{\frac{\sum_{(p,g)\in \mathrm{TP}} \mathrm{IoU}(p,g)}{|\mathrm{TP}|}}_{\text{Segmentation Quality (SQ)}} \times \underbrace{\frac{|\mathrm{TP}|}{|\mathrm{TP}| + \frac{1}{2}|\mathrm{FP}| + \frac{1}{2}|\mathrm{FN}|}}_{\text{Recognition Quality (RQ)}}$$

so it reflects overlap quality for matched instances while also penalising missing and false objects. This makes PQ less forgiving than pixel-level Dice because missing and false instances are penalised directly (Kirillov et al., 2019).

Compared with Dice or IoU, these metrics ask a harder and more relevant question: whether the segmentation recovers the right mitochondria as distinct biological objects and preserves separations between them. They prove useful when two models appear similar on overlap metrics but differ in how often they merge or fragment neighbouring organelles. This makes them well suited to the main comparison in this dissertation.

Some of the object metrics used in this work are more diagnostic than others. Matched instance IoU measures how well the predicted objects that do correspond to real mitochondria overlap their targets once matching has been established. It separates the quality of matched objects from the problem of missing or extra objects. Object precision and object recall at multiple IoU thresholds make this looser or stricter in a controlled way: low thresholds ask whether the right objects were found at all, while high thresholds ask whether they were recovered accurately enough for detailed biological analysis.

These metrics also have limitations. Count error is easy to interpret, but it does not explain why the count is wrong. Merge and split rates are highly diagnostic, but less standardised across papers than Dice or IoU. PQ is less forgiving than Dice, but it still depends on a matching threshold and compresses multiple behaviours into one number. The implication is not that any single object metric solves evaluation on its own. Object-level metrics provide the missing evidence needed to interpret whether strong semantic scores actually correspond to biologically reliable predictions (Wei et al., 2020; Conrad and Narayan, 2023).

3.4 Boundary and Calibration Metrics

Between pure overlap metrics and pure object-level metrics, two intermediate families of measures are also useful: boundary quality and calibration. They show different weaknesses in model behaviour (Taha and Hanbury, 2015; Guo et al., 2017).

HD95 captures large boundary deviations while reducing sensitivity to single extreme outliers, which makes it useful for detecting contour errors that Dice or IoU can miss

(Taha and Hanbury, 2015). Boundary F1 (BF1) measures how much of the predicted and true boundary lies within a chosen tolerance band (Martin et al., 2001). In this work, 2px and 5px are used as stricter and more forgiving views of boundary accuracy. These metrics are more sensitive to boundary quality than overlap scores alone.

Expected Calibration Error (ECE) measures whether predicted confidence matches observed accuracy (Guo et al., 2017). This matters because a model can achieve strong overlap while remaining overconfident or underconfident, which makes threshold choice and uncertainty interpretation less reliable.

3.5 The Full Metric Set Used in This Dissertation

The evaluation therefore combines several metric groups. A single headline score would not answer the dissertation's questions adequately. The full evaluated set is chosen to answer the dissertation's three aims: compare semantic overlap, assess boundary and confidence quality, and expose object-level biological failures that Dice alone can hide.

- **Pixel overlap:** foreground Dice, background Dice, mean Dice, IoU, precision, and recall. These are included because they remain the standard semantic-segmentation summaries in the literature.
- **Boundary quality:** HD95, boundary F1 at 2 pixels, and boundary F1 at 5 pixels. These are included because biologically relevant errors can arise from misplaced or coarse boundaries even when pixel overlap remains high.
- **Calibration and threshold robustness:** ECE and foreground Dice at thresholds 0.3, 0.4, 0.5, 0.6, and 0.7. ECE tests whether predicted probabilities are trustworthy, while the threshold analysis checks whether the apparent model ranking depends strongly on a single operating point. The threshold is the prediction cutoff used to binarise the probability map before Dice is recomputed.
- **Object recovery and topology:** GT objects per image, predicted objects per image, absolute count error, signed count error, merge events, split events, merge rate, split rate, connectivity error, mean matched IoU, Panoptic Quality at IoU > 0.5, and object precision/recall at IoU thresholds 0.3, 0.5, 0.7, and 0.9. These are included because the biology-facing task depends on recovering distinct mitochondria as objects. The threshold is the object-match IoU criterion used after object recovery to decide whether a predicted object counts as correctly predicted.
- **Failure diagnostics beyond scalar metrics:** zero-Dice slice counts, completely missed GT-positive slices, false positives on GT-empty slices, the high-Dice corner-case diagnostic, and stratified summaries by foreground-density quartile. These are included so that specific failure patterns remain visible in the analysis.

The full evaluated set is broader than the subset of 27 paired per-image metrics used for formal paired statistical testing. This inferential subset was chosen from the per-image metrics that are naturally paired across the two models and that cover the main evaluation

families: pixel overlap, boundary quality, calibration, threshold sensitivity, and object-level behaviour. Other reported quantities are treated as descriptive diagnostics.

3.6 Gap Addressed by This Dissertation

The literature establishes that U-Net-style semantic models remain widely used and that nnU-Net is a strong self-configuring baseline (Ronneberger et al., 2015; Isensee et al., 2021). It also shows that overlap metrics such as Dice remain the dominant reporting standard and may be insufficient for biology-facing evaluation (Franco-Barranco et al., 2022; Wei et al., 2020; Conrad and Narayan, 2023). What is still missing is a controlled comparison between semantic models under a shared evaluation setup.

Across the papers discussed above, model comparisons are often made under different supervision setups, datasets, dimensionalities, or evaluation protocols (Wei et al., 2020; Conrad and Narayan, 2023; Xie et al., 2025; Chen et al., 2025). As a result, it is often hard to separate differences in model behaviour from differences in experimental setup. This dissertation addresses that gap by comparing a custom U-Net and nnU-Net on the same biological source-level split, with shared post-processing and both pixel-level and object-level evaluation on CEM-MitoLab. The aim is to test whether overlap-based evaluation alone gives a complete account of model quality in this setting.

Chapter 4

Methodology and Implementation

4.1 Experiment Setup

This dissertation uses a controlled comparative experiment. The U-Net and nnU-Net share the same source-level split, the same held-out test set, and the same object-recovery and evaluation procedure, while model-specific internals are allowed to differ.

The experimental setup is defined first at the level of the data split. Although CEM-MitoLab consists of individual 2D EM slices, these slices come from larger biological sources. A *biological source* is a single contributing entry in the dataset: one cell, tissue volume, or imaging session from which multiple slices were taken. Because slices from the same source share imaging conditions and annotation style, splitting at the slice level would risk placing near-identical images in both training and test. The train, validation, and test sets are therefore defined at the source level and then expanded to slice level for model training and evaluation.

Both models are trained as semantic segmenters on binary foreground masks, so the task is the same in each pipeline. They identify mitochondria pixels against background. Evaluation then proceeds in two stages. First, the models are assessed as pixel-wise foreground predictors using pixel-level, boundary, and calibration metrics. Second, their binary outputs are converted into recovered object sets by connected-component labelling, so that the same predictions can also be assessed at the level of individual mitochondria.

Other factors remain model-specific, including architecture, batch size, optimiser behaviour, and nnU-Net's automatically selected configuration. The aim is not to make the pipelines identical. Instead, the comparison is controlled closely enough that observed differences can be interpreted as model behaviour, not as artifacts of data handling or evaluation.

4.2 System Pipeline

At a high level, the pipeline consists of shared data preparation, two model-specific training and inference branches, and a common evaluation stage. The shared infrastructure written for this project includes dataset conversion, split synchronisation, prediction export, and the evaluation code. Separate custom code was also developed for U-Net training and inference, while nnU-Net itself was used as an external package without modification to its core training logic. This is shown in Figure 4.1. The main implementation effort therefore lay in building a consistent comparison pipeline around the two models.

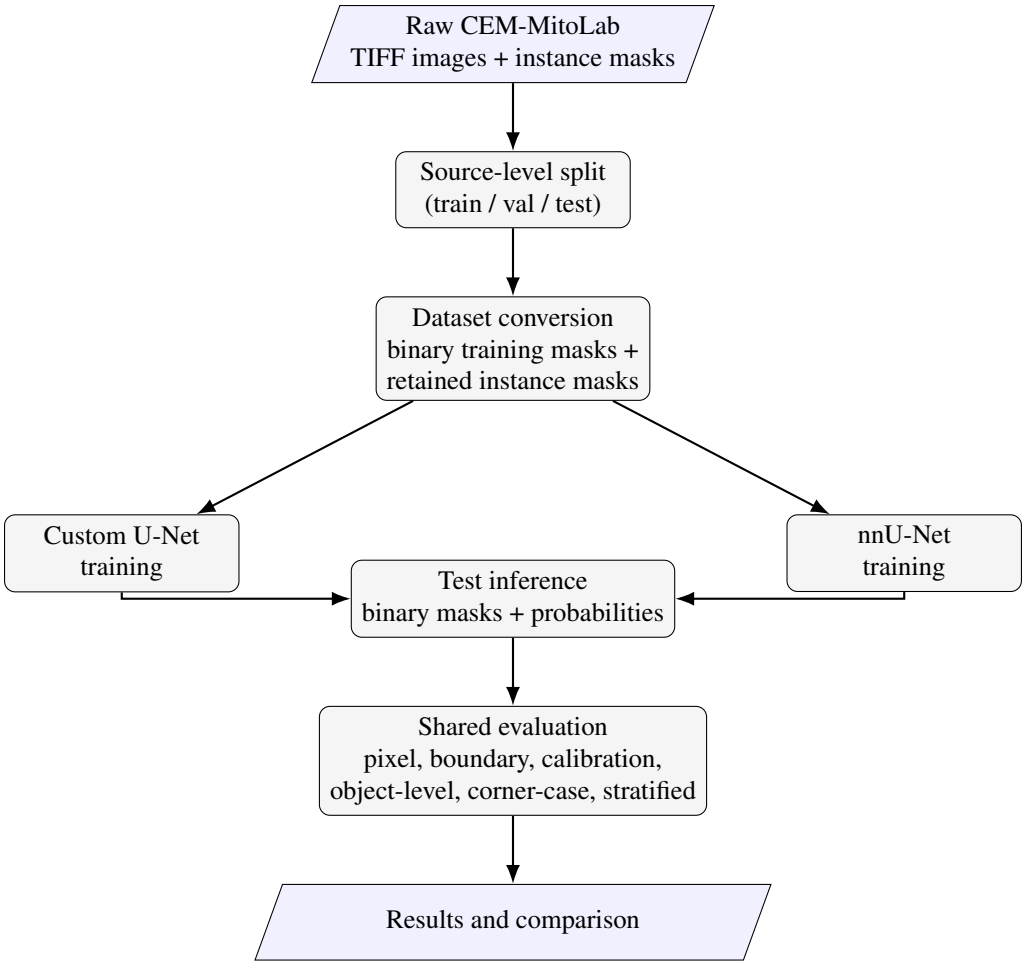


Figure 4.1: Overview of the experimental pipeline. The two models share the same source-level split and a common evaluation stage.

4.3 Dataset Description and Preprocessing

CEM-MitoLab consists of 2D electron microscopy slices with instance-labelled mitochondria masks. For the present experiment, the unprocessed images and annotations were converted into a common format suitable for both model training and shared evaluation. Binary semantic masks were derived from the instance labels for training,

while the original instance masks were retained for later object-level evaluation.

Table 4.1 summarises the final split used throughout the dissertation.

Table 4.1: Final source-level and slice-level split used for all experiments.

Partition	Sources	Slices	Role
Train	415	15,312	Model fitting and augmentation
Validation	89	3,247	Model selection and early stopping
Test	89	3,312	Final test set evaluation only

The same split definition was used consistently for both models throughout training, validation, and final testing. During training and validation, both models used the same preprocessed inputs. For final inference, both were applied to the same unseen unprocessed test images before entering the shared evaluation pipeline.

The image slices varied in spatial dimensions and were therefore kept at their original resolution rather than resized to a common size. This preserved the full field of view and avoided altering object scale through resampling. For the custom U-Net, images were padded only when required for processing, and predictions were then returned to the original image size. For nnU-Net, the images were converted directly into the required input format and processed using the framework's standard procedures. This avoided manual cropping or rescaling and helped ensure that the comparison reflected model behaviour rather than differences introduced by additional preprocessing.

4.4 Model Configurations

4.4.1 Custom U-Net

The custom model is a 2D U-Net implemented in PyTorch from scratch. It follows the standard encoder-decoder design with skip connections, using four down-sampling blocks and a bottleneck, with channel widths 64, 128, 256, 512, and 1024 before the decoder upsamples back to full resolution. Each convolutional block applies two 3×3 convolutions followed by instance normalisation and rectified linear unit (ReLU) activations (Nair and Hinton, 2010). Instance normalisation was chosen because the custom model trains with a batch size of 4, for which batch-normalisation statistics are unstable.

To align the U-Net training objective with nnU-Net's standard formulation, the model predicts two classes (background and mitochondria) and is trained with a combined cross-entropy plus soft Dice objective. For a training batch of N pixels, with class probability $p_{i,c}$ and target label $y_{i,c}$ for pixel i and class c, the cross-entropy term is

$$\mathcal{L}_{\mathrm{CE}} = -\frac{1}{N}\sum_{i=1}^{N}\sum_{c} y_{i,c}\log p_{i,c}.$$

For the foreground class alone, the soft Dice loss is

$$\mathcal{L}_{\text{Dice}} = -\frac{2\sum_{i=1}^{N} p_i y_i + \varepsilon}{\sum_{i=1}^{N} p_i + \sum_{i=1}^{N} y_i + \varepsilon},$$

where p_i and y_i denote the predicted foreground probability and the binary foreground label, respectively. The total training objective is

$$\mathcal{L} = \mathcal{L}_{\text{CE}} + \mathcal{L}_{\text{Dice}}.$$

This combined objective follows the standard nnU-Net formulation and the common soft Dice approach used in biomedical segmentation (Isensee et al., 2021; Milletari et al., 2016). The Dice term is applied only to the foreground class. The reported training and validation Dice is computed from epoch-level hard true-positive, false-positive, and false-negative counts.

4.4.2 nnU-Net

nnU-Net was used in its standard form, without changing the way it trains. For this dataset it selected the 2D configuration, with a patch size of 512×512 and batch size 12. The resulting network is a plain convolutional U-Net with eight stages, feature widths {32, 64, 128, 256, 512, 512, 512, 512}, instance normalisation, and leaky rectified linear unit (LeakyReLU) activations (Maas et al., 2013). These choices come directly from nnU-Net's planning stage and define the external baseline used in the comparison.

Table 4.2 summarises the main design and training differences between the two systems.

Table 4.2: Summary of the two model setups used in the comparison.

Feature	**Custom U-Net**	**nnU-Net**
Task	2D semantic segmentation	2D semantic segmentation
Train/val input	Same preprocessed training and validation images	Same preprocessed training and validation images
Architecture	Four-level U-Net with skip connections	Automatically configured 2D convolutional U-Net
Channels/stages	64, 128, 256, 512, 1024	32, 64, 128, 256, 512, 512, 512, 512
Normalisation	Instance normalisation	Instance normalisation
Activation	Rectified linear unit activations	Leaky rectified linear unit activations
Loss	Combined cross-entropy and soft Dice objective	Combined cross-entropy and soft Dice objective
Batch size	4	12
Probability export	Binary masks and probability maps exported for shared evaluation	Binary masks and probability maps exported for shared evaluation

4.5 Training and Inference Procedure

4.5.1 Software and Compute Environment

The custom pipeline was implemented in Python using PyTorch `2.8.0+cu128`, with GPU execution through CUDA under SLURM. nnU-Net was used through its v2.6 command-line interface as an external package, and its core code was left unchanged. Both training pipelines were run as single-GPU jobs, primarily on an NVIDIA GeForce RTX 2080 Ti (11 GB VRAM), with 32 GB RAM and two CPU workers.

4.5.2 Training Procedure

Both models were trained on the same fixed training and validation split, and model selection was validation-based. During training, each pipeline tracked a Dice-style validation summary to monitor progress. This quantity was used only for training-time monitoring and model selection, not as a final test-set metric.

4.5.2.1 U-Net

The custom U-Net was trained with the AdamW optimiser (Loshchilov and Hutter, 2019) and an initial learning rate of 10^{-3}. The maximum schedule was 1,000 epochs. Validation was performed every second epoch, and the learning rate was reduced when validation performance stopped improving. Training was stopped early if the validation loss failed to improve meaningfully over 50 validation checks.

Data augmentation was applied only to the training split. This consisted of horizontal and vertical flips, 90-degree rotations, small brightness shifts, contrast changes, and additive Gaussian noise.

4.5.2.2 nnU-Net

nnU-Net was trained in its standard configuration for this dataset for the full 1,000-epoch schedule. It was trained on fold 0 only. The five-fold ensemble was not used because the aim was a fair single-model comparison under the same split and a similar training budget. Its per-epoch validation log reports pseudo-Dice, while checkpoint selection uses the exponential moving average (EMA pseudo-Dice) (Isensee et al., 2021). Validation-time probability outputs were retained during training, and the final test-time probability outputs were exported separately during inference for the later calibration and threshold-sensitivity analyses.

4.5.3 Model Prediction on the Test Set

For the custom U-Net, the best validation checkpoint is applied to the test cases. The shared default threshold of 0.5 is then used to binarise the foreground probability map, and both binary masks and compressed probability arrays are exported. U-Net test inference uses unprocessed TIFF images with z-score normalisation to match training as closely as possible.

For nnU-Net, the selected test-time checkpoint is applied to the same test images, and both binary masks and probability maps are retained for the shared evaluation stage.

4.6 Applied Evaluation Procedure

The metric definitions and rationale are given in Chapter 3. This section describes how that framework was operationalised in the present experiment. For each test case, the shared evaluation procedure uses a binary prediction mask, a foreground probability map, and the original instance-labelled ground truth. The same post-processing and measurement steps are applied to both models.

4.6.1 Case Grouping and Aggregation

All evaluation is carried out on the same set of 3,312 test slices. Because 618 of these are background-only, aggregate results are reported both over the full test set and over the 2,694 GT-positive slices that contain at least one mitochondrion in the ground truth. This distinction is important because empty slices make some predictions easy and would otherwise blur differences in foreground and object-level behaviour. For foreground-only overlap metrics, empty-empty slices are excluded automatically.

4.6.2 Pixel-Level Metrics

Pixel-level metrics are computed at foreground thresholds $\{0.3, 0.4, 0.5, 0.6, 0.7\}$. At each threshold the evaluation records foreground Dice, background Dice, mean Dice, IoU, precision, recall, HD95, and boundary F1 at 2-pixel and 5-pixel tolerance. HD95 is computed from the distances between predicted and ground-truth boundary pixels, using Euclidean distance transforms. The calculation is therefore based on boundary pixels, not all pixels in the mask. For HD95 specifically, empty-empty slices are assigned HD95 = 0, whereas slices where exactly one mask is empty are treated as non-finite and excluded from finite-case summaries. Expected Calibration Error (ECE) is computed from the probability maps using 15 bins. Threshold 0.5 is treated as the default operating point, and the surrounding thresholds are used for the threshold-sensitivity analysis.

4.6.3 Object-Level Metrics

Predicted instances are recovered from each binary foreground mask by connected-component labelling with a 3×3 connectivity structure. In the final reported analysis, all predicted components are retained before matching. The predicted objects are then compared with the ground-truth instance masks by building a full pairwise IoU matrix and applying Hungarian matching to obtain one-to-one assignments (Kuhn, 1955; Munkres, 1957). This produces absolute count error, signed count error, mean matched IoU (Taha and Hanbury, 2015), Panoptic Quality at IoU > 0.5 (Kirillov et al., 2019), object-level precision and recall at IoU thresholds 0.3, 0.5, 0.7, and 0.9, and the merge, split, and connectivity statistics. A predicted object overlapping multiple ground-truth mitochondria is counted as a merge, whereas a ground-truth mitochondrion overlapped by multiple predicted objects is counted as a split.

4.6.4 High-Dice Corner Cases

The evaluation also includes a corner-case diagnostic. A *high-Dice corner case* is defined as a test slice with foreground Dice above 0.85 at the default threshold of 0.5, but with either object count error greater than 4, merge rate greater than 0, or split rate greater than 0. All high-Dice slices in the evaluation are GT-positive, because empty-empty slices are excluded from foreground-Dice summaries. The rule is used here as a descriptive filter for slices that already achieve strong overlap. Prior work has shown that high overlap can coexist with biologically important structural failures (Taha and Hanbury, 2015; Franco-Barranco et al., 2022; Wei et al., 2020; Conrad and Narayan, 2023). The Dice cutoff of 0.85 was chosen to focus the diagnostic on the upper score range while retaining enough cases for meaningful analysis in both models. The object-level side of the rule treats merge and split as strict binary conditions, because either event changes object identity and separation regardless of its magnitude. The count error threshold being greater than 4 was chosen to target the upper tail of the GT-positive count-error distribution (75th percentile: 5 for U-Net, 4 for nnU-Net), as shown in Appendix Figure A.1. No universal threshold exists for this kind of diagnostic, and these values are choices, but they are grounded in the observed error distribution.

4.6.5 Statistical Testing

Direct model comparisons are performed on paired per-image results. The shared case identifiers in the paired per-image result tables are aligned first, after which each metric is tested with a two-sided Wilcoxon signed-rank test (Wilcoxon, 1945). Pairs containing `NaN` or infinite values are removed on a per-metric basis so that each test is run only where the relevant metric is defined for both models.

Formal inference is applied to the set of 27 paired per-image metrics described in Chapter 3, spanning pixel-level overlap, boundary, calibration, threshold sensitivity, and object-level behaviour. Descriptive quantities such as signed count error, raw merge and split event counts, zero-Dice counts, missed-positive counts, GT-empty false positives, corner-case rates, and stratified summaries are reported separately and are not included in the formal inference. Raw p-values are corrected using the Benjamini–Hochberg false-discovery-rate procedure (Benjamini and Hochberg, 1995) with $\alpha = 0.05$.

4.6.6 Stratified Analysis

The main stratified analysis is included to check whether the overall test-set comparison is consistent across different image conditions, or whether it is being driven by a particular subset of slices. GT-positive test slices are grouped by foreground density, measured as the proportion of image pixels labelled as mitochondria in the ground truth. The ground-truth foreground-ratio distribution is divided into four quartiles, and the same quartile boundaries are then applied to both models, so the grouping is fixed independently of the predictions. This yields four strata of roughly equal size, each containing about 674 slices. Within each stratum, the models are compared using the same key metrics as in the main analysis, particularly foreground Dice, object count error, split rate, and Panoptic Quality.

Chapter 5

Results and Analysis

5.1 Test-Set Overview

Within this fixed test set, 2,694 images (81.3%) contain at least one mitochondrion in the ground truth (*GT-positive*), while the remaining 618 (18.7%) are background-only slices. Because background-only images allow any model to achieve trivially perfect background Dice and an object count of zero by predicting nothing, the distinction between GT-positive and GT-empty cases is maintained throughout the analysis. Unless otherwise stated, all aggregate figures are reported over all 3,312 test images; supplementary GT-positive-only figures are provided where the difference is meaningful.

5.2 U-Net Evaluation

5.2.1 Pixel-Level Performance

Table 5.1 summarises the final U-Net pixel-level metrics at the default threshold of 0.5.

Table 5.1: U-Net pixel-level metrics at threshold 0.5 on the test set. *All*: aggregate over the 3,312-case test set; *GT+*: GT-positive cases only (2,694 cases). Foreground-only overlap metrics exclude empty-empty slices, where foreground overlap is undefined. HD95 is reported over finite cases only.

	All		**GT+**	
Metric	Mean ± SD	Median [IQR]	Mean ± SD	Median [IQR]
Dice FG	0.578 ± 0.351	0.714 [0.262–0.879]	0.612 ± 0.331	0.735 [0.423–0.885]
Dice BG	0.985 ± 0.024	0.993 [0.981–0.998]	0.983 ± 0.025	0.990 [0.978–0.996]
Dice Mean	0.780 ± 0.177	0.845 [0.616–0.933]	0.797 ± 0.167	0.858 [0.696–0.936]
IoU (FG)	0.485 ± 0.325	0.555 [0.151–0.784]	0.514 ± 0.312	0.582 [0.269–0.794]
Precision	0.671 ± 0.375	0.876 [0.463–0.937]	0.711 ± 0.347	0.885 [0.614–0.941]
Recall	0.558 ± 0.362	0.652 [0.205–0.891]	0.591 ± 0.346	0.686 [0.328–0.899]
BF1 (2 px)	0.626 ± 0.355	0.748 [0.386–0.915]	0.599 ± 0.328	0.712 [0.392–0.859]
BF1 (5 px)	0.681 ± 0.362	0.839 [0.491–0.973]	0.667 ± 0.340	0.804 [0.496–0.929]
HD95 (finite)	47.2 ± 68.8	18.5 [2.2–63.1]	56.1 ± 71.7	29.9 [6.7–78.6]
ECE	0.020 ± 0.033	0.009 [0.002–0.025]	0.023 ± 0.035	0.013 [0.005–0.028]

At threshold 0.5, the U-Net mean foreground Dice reaches 0.578 over all slices where foreground overlap is defined and 0.612 over GT-positive cases only, with mean IoU at 0.485. The medians are consistently higher than the means. Most slices perform better than the average, but a minority of hard cases pulls it down. Precision (0.671) exceeds recall (0.558), which reflects a conservative foreground bias. Missed foreground often corresponds to mitochondria that are only partially recovered, or missed entirely.

HD95 is undefined on 438 images (13.2%). Over the remaining 2,874 finite cases, the mean HD95 is 47.2 px and the median is 18.5 px. For U-Net, the 438 non-finite cases therefore correspond to 278 completely missed GT-positive slices and 160 GT-empty false positives. The gap between the mean and median indicates that most predictions are much closer to the boundary than the mean alone suggests. The mean is pulled upward by a smaller set of outliers with large contour errors.

The U-Net gives a mean ECE of 0.020 and a median of 0.009, indicating that its foreground probabilities are generally close to the observed fraction of foreground pixels.

5.2.2 Threshold Sensitivity

Table 5.2 reports U-Net pixel metrics at five probability thresholds spanning the range 0.30–0.70. Overlap and boundary metrics are reported as mean ± standard deviation across the 3,312 test cases, while HD95 is reported over the finite cases at each threshold.

Table 5.2: U-Net pixel metrics at five probability thresholds. Foreground-only overlap and boundary metrics are reported as mean ± standard deviation over the cases where they are defined; HD95 is also reported as mean ± standard deviation over finite cases only.

Metric	$t = 0.30$	$t = 0.40$	$t = 0.50$	$t = 0.60$	$t = 0.70$
Dice FG	0.583 ± 0.352	0.581 ± 0.351	0.578 ± 0.351	0.574 ± 0.350	0.570 ± 0.350
Dice BG	0.985 ± 0.024	0.985 ± 0.024	0.985 ± 0.024	0.985 ± 0.024	0.985 ± 0.024
Dice Mean	0.783 ± 0.177	0.782 ± 0.177	0.780 ± 0.177	0.779 ± 0.177	0.776 ± 0.177
IoU (FG)	0.492 ± 0.326	0.489 ± 0.326	0.485 ± 0.325	0.482 ± 0.325	0.477 ± 0.324
Precision	0.656 ± 0.370	0.664 ± 0.373	0.671 ± 0.375	0.675 ± 0.378	0.681 ± 0.380
Recall	0.577 ± 0.368	0.567 ± 0.365	0.558 ± 0.362	0.549 ± 0.360	0.538 ± 0.357
BF1 (2 px)	0.627 ± 0.355	0.627 ± 0.355	0.626 ± 0.355	0.625 ± 0.355	0.622 ± 0.356
BF1 (5 px)	0.682 ± 0.363	0.681 ± 0.362	0.681 ± 0.362	0.679 ± 0.362	0.677 ± 0.362
HD95	46.9 ± 68.5	46.9 ± 68.4	47.2 ± 68.8	47.2 ± 68.2	47.6 ± 68.1

Foreground Dice varies by only 0.013 across thresholds 0.30–0.70 (from 0.583 to 0.570), with precision rising slightly and recall falling as the threshold increases. Finite-case HD95 also remains broadly stable across the same range. This highlights how the threshold choice is not a major source of variation for the U-Net performance.

5.2.3 Object-Level and Biological Performance

Table 5.3 summarises the U-Net object-level metrics under the connected-component recovery and Hungarian matching procedure.

Table 5.3: U-Net object-level metrics under the connected-component analysis, with all predicted components retained before matching. *All*: 3,312 test cases; *GT+*: GT-positive cases only (2,694 cases).

	All		**GT+**	
Metric	Mean ± SD	Median [IQR]	Mean ± SD	Median [IQR]
GT objs/image	6.57 ± 7.14	3 [1–11]	8.08 ± 7.11	5 [2–13]
Pred. objs/image	6.61 ± 7.27	4 [1–11]	7.89 ± 7.35	6 [2–13]
Abs. count error	3.07 ± 4.52	2 [0–4]	3.54 ± 4.68	2 [1–5]
Signed count error	0.04 ± 5.46	0 [−2–1]	-0.18 ± 5.86	0 [−3–1]
Merge rate	0.041 ± 0.099	0.000	0.050 ± 0.108	0.000 [0.000–0.041]
Split rate	0.091 ± 0.320	0.000	0.112 ± 0.352	0.000 [0.000–0.091]
Conn. error	0.132 ± 0.346	0.000 [0.000–0.151]	0.162 ± 0.378	0.000 [0.000–0.222]
Matched IoU	0.482 ± 0.355	0.625 [0.000–0.788]	0.592 ± 0.299	0.691 [0.499–0.813]
PQ (IoU > 0.5)	0.459 ± 0.340	0.445 [0.170–0.715]	0.395 ± 0.275	0.403 [0.171–0.603]

The object-level metrics present a more constrained view of performance than the pixel-level summary. PQ reaches 0.459 over all test cases and 0.395 over GT-positive

cases only, while mean matched IoU is 0.482 over all cases. The near-zero signed count error is misleading, because missed objects and extra fragments partly cancel when averaged across slices.

The average count error remains substantial at 3.07 over all test cases and 3.54 on GT-positive slices, with a median of 2 in both views. On GT-positive slices the signed count error is still slightly negative (-0.18). The bias is toward under-estimating the number of mitochondria present in each slice, although fragmented over-prediction in other slices shifts the overall mean closer to zero.

The median merge and split rates are both zero, indicating that many images do not exhibit topology errors. The mean split rate of 0.091 and connectivity error of 0.132 show, however, that fragmentation remains concentrated in a subset of difficult cases. Most images are unaffected, but the failures that do occur are large enough to raise the mean.

Table 5.4 reports object-level precision and recall at four IoU thresholds for all test cases.

Table 5.4: U-Net object-level precision and recall at four IoU thresholds under the same connected-component analysis (mean $\pm$ SD over all 3,312 test cases).

IoU thr.	≥ 0.3	≥ 0.5	≥ 0.7	≥ 0.9
Obj. Precision	0.594 ± 0.381	0.557 ± 0.382	0.492 ± 0.382	0.247 ± 0.366
Obj. Recall	0.593 ± 0.374	0.558 ± 0.379	0.494 ± 0.384	0.246 ± 0.367

At the lenient IoU threshold of 0.3, object-level precision reaches 0.594 and recall 0.593. These fall to 0.557 and 0.558 at IoU 0.5 and then collapse to 0.247 and 0.246 at the strict threshold of 0.9. The model often places mitochondria in roughly the right location. It is less reliable when considering whether those mitochondria were separated correctly.

5.2.4 Failure Pattern

The U-Net produces 551 zero-Dice slices, 278 completely missed GT-positive slices, and 160 false-positive predictions on GT-empty slices. These counts point to two recurring failure modes: complete misses on positive images and false detections on background-only slices. The zero-Dice cases occur mainly in slices with very small foreground extent, weak local contrast, or fragmented predictions that fail to overlap the annotated mitochondria sufficiently. The false positives on GT-empty slices show that fine background texture or local noise is sometimes converted into small foreground fragments when all predicted components are retained. In biological applications, these two errors have different consequences. Missed positive slices lead to under-estimation of mitochondrial presence or density, whereas false positives on empty slices inflate counts in regions that should contribute no mitochondria at all. As a result, these failure modes account for much of the low-performing tail in the pixel-level distribution and help explain why the mean is notably lower than the median. This interpretation is

consistent with the stratified analysis in Section 5.5, where the low-density foreground quartile shows the weakest pixel-level and object-level results.

5.3 nnU-Net Evaluation

5.3.1 Pixel-Level Performance

Table 5.5 summarises the pixel-level metrics at the default prediction threshold of 0.5, for all test cases and for GT-positive cases only. All values report mean ± standard deviation and the median with interquartile range.

Table 5.5: nnU-Net pixel-level metrics at threshold 0.5 on the test set. *All*: aggregate over the 3,312-case test set; *GT+*: GT-positive cases only (2,694 cases). Foreground-only overlap metrics exclude empty-empty slices, where foreground overlap is undefined. HD95 is reported over finite cases only.

	All		**GT+**	
Metric	Mean ± SD	Median [IQR]	Mean ± SD	Median [IQR]
Dice FG	0.624 ± 0.353	$0.792\ [0.366-0.907]$	0.643 ± 0.342	$0.804\ [0.441-0.909]$
Dice BG	0.987 ± 0.025	$0.995\ [0.986-0.999]$	0.985 ± 0.027	$0.993\ [0.984-0.997]$
Dice Mean	0.805 ± 0.180	$0.890\ [0.671-0.949]$	0.814 ± 0.174	$0.894\ [0.706-0.950]$
IoU (FG)	0.539 ± 0.335	$0.655\ [0.224-0.830]$	0.554 ± 0.326	$0.673\ [0.283-0.833]$
Precision	0.729 ± 0.367	$0.914\ [0.712-0.961]$	0.750 ± 0.350	$0.917\ [0.762-0.963]$
Recall	0.593 ± 0.363	$0.740\ [0.263-0.910]$	0.610 ± 0.354	$0.756\ [0.328-0.913]$
BF1 (2 px)	0.684 ± 0.354	$0.837\ [0.516-0.967]$	0.641 ± 0.344	$0.775\ [0.435-0.909]$
BF1 (5 px)	0.729 ± 0.354	$0.901\ [0.627-1.000]$	0.696 ± 0.350	$0.853\ [0.566-0.954]$
HD95 (finite)	40.3 ± 65.7	$8.9\ [1.4-55.4]$	49.6 ± 69.6	$21.0\ [3.6-66.9]$
ECE	0.018 ± 0.038	$0.007\ [0.001-0.018]$	0.022 ± 0.040	$0.010\ [0.004-0.022]$

At threshold 0.5, nnU-Net reaches a mean foreground Dice of 0.624 over all usable slices and 0.643 over GT-positive cases only. As with the U-Net, the median substantially exceeds the mean, so a minority of difficult images still drives much of the remaining error. Precision again exceeds recall (0.729 vs. 0.593). The model is still more prone to missing foreground than to adding it, and that shows up later as object under-counting.

HD95 is non-finite on 409 test images (12.3%). Among the remaining 2,903 finite cases, the mean HD95 is 40.3 px (SD 65.7), and the median is 8.9 px. This indicates that most predictions lie close to the ground-truth boundary, with a smaller subset exhibiting large localised deviations. For GT-positive cases only (2,363 finite), the mean rises to 49.6 px and the median to 21.0 px, as the easy background-only cases no longer contribute. In this case, empty-empty slices receive HD95 = 0, so the 409 non-finite nnU-Net cases correspond specifically to 331 missed GT-positive slices and 78 GT-empty false positives.

The mean ECE of 0.018 (median 0.007) indicates well-calibrated probability maps overall. The low median relative to the mean again reflects a small proportion of poorly calibrated images.

5.3.2 Threshold Sensitivity

Table 5.6: nnU-Net pixel metrics at five probability thresholds. Foreground-only overlap and boundary metrics are reported as mean ± standard deviation over the cases where they are defined; HD95 is also reported as mean ± standard deviation over finite cases only.

Metric	$t = 0.30$	$t = 0.40$	$t = 0.50$	$t = 0.60$	$t = 0.70$
Dice FG	0.635 ± 0.351	0.629 ± 0.353	0.624 ± 0.353	0.618 ± 0.355	0.611 ± 0.356
Dice BG	0.988 ± 0.025	0.988 ± 0.025	0.987 ± 0.025	0.987 ± 0.025	0.987 ± 0.025
Dice Mean	0.810 ± 0.178	0.807 ± 0.179	0.805 ± 0.180	0.801 ± 0.180	0.798 ± 0.181
IoU (FG)	0.549 ± 0.333	0.544 ± 0.334	0.539 ± 0.335	0.532 ± 0.335	0.525 ± 0.335
Precision	0.720 ± 0.361	0.724 ± 0.364	0.729 ± 0.367	0.732 ± 0.370	0.735 ± 0.374
Recall	0.614 ± 0.365	0.603 ± 0.364	0.593 ± 0.363	0.581 ± 0.362	0.569 ± 0.360
BF1 (2 px)	0.690 ± 0.351	0.687 ± 0.353	0.684 ± 0.354	0.680 ± 0.356	0.675 ± 0.357
BF1 (5 px)	0.734 ± 0.353	0.731 ± 0.354	0.729 ± 0.354	0.725 ± 0.356	0.720 ± 0.358
HD95	40.9 ± 66.5	40.6 ± 65.7	40.3 ± 65.7	40.8 ± 66.1	41.4 ± 66.5

Foreground Dice varies by only 0.024 across thresholds 0.30–0.70 (from 0.635 to 0.611), and HD95 is essentially stable across this range. Threshold choice is therefore not a major source of variation for nnU-Net.

5.3.3 Object-Level and Biological Performance

Table 5.7 summarises the object-level metrics under the same connected-component recovery used for the U-Net. Ground-truth objects are taken from the original instance masks provided with the dataset, where background is labelled 0 and individual mitochondria are represented by nonzero integer labels.

Table 5.7: nnU-Net object-level metrics under the connected-component analysis, with all predicted components retained before matching. *All*: 3,312 test cases; *GT+*: GT-positive cases only (2,694 cases).

	All		**GT+**	
Metric	Mean ± SD	Median [IQR]	Mean ± SD	Median [IQR]
GT objs/image	6.57 ± 7.14	3 [1–11]	8.08 ± 7.11	5 [2–13]
Pred. objs/image	5.13 ± 5.69	3 [0–9]	6.23 ± 5.72	4 [2–10]
Abs. count error	2.56 ± 3.70	1 [0–3]	3.07 ± 3.86	2 [1–4]
Signed count error	-1.44 ± 4.26	0 [−3–0]	-1.85 ± 4.57	−1 [−4–0]
Merge rate	0.035 ± 0.090	0.000	0.044 ± 0.098	0.000
Split rate	0.041 ± 0.266	0.000	0.050 ± 0.294	0.000
Conn. error	0.076 ± 0.282	0.000 [0.000–0.053]	0.093 ± 0.310	0.000 [0.000–0.100]
Matched IoU	0.514 ± 0.373	0.698 [0.000–0.826]	0.632 ± 0.310	0.750 [0.580–0.846]
PQ (IoU > 0.5)	0.544 ± 0.341	0.577 [0.299–0.831]	0.468 ± 0.295	0.520 [0.247–0.702]

The signed count error of -1.44 overall and -1.85 on GT-positive slices confirms systematic under-prediction of objects. This is consistent with the pixel-level recall gap (0.593 vs. precision 0.729), where the model misses some mitochondria even when it overlaps the foreground well at the pixel level.

Merge rate (0.035) and split rate (0.041) have low mean values, and the median for both is exactly zero, indicating that most images have no merge or split events. Relative to the U-Net, both means are lower, especially for split rate (0.041 vs. 0.091), so fragmentation is less frequent. The remaining errors are still concentrated in a subset of difficult cases. They are not spread evenly across the test set.

Table 5.8 reports object-level precision and recall at four IoU thresholds for all test cases.

Table 5.8: nnU-Net object-level precision and recall at four IoU thresholds under the same connected-component analysis (mean ± SD over all 3,312 test cases).

IoU thr.	≥ 0.3	≥ 0.5	≥ 0.7	≥ 0.9
Obj. Precision	0.709 ± 0.375	0.676 ± 0.377	0.611 ± 0.382	0.329 ± 0.390
Obj. Recall	0.633 ± 0.367	0.606 ± 0.372	0.556 ± 0.378	0.308 ± 0.387

At the lenient IoU threshold of 0.3, object-level precision reaches 0.709 and recall 0.633. These fall to 0.676 and 0.606 at IoU 0.5 and decline further to 0.329 and 0.308 at the threshold of 0.9. Many mitochondria are detected at the right location, but fewer are recovered with boundaries precise enough for strict object-level agreement. PQ at IoU > 0.5 (mean 0.544, median 0.577) reflects both detection and localisation.

5.3.4 Failure Pattern

nnU-Net's failure profile differs from U-Net's in a consistent way: it misses more GT-positive slices outright (331 vs. 278) but produces far fewer false positives on GT-empty slices (78 vs. 160). The model is more conservative. Where U-Net converts background texture or weak signal into small foreground fragments, nnU-Net more often predicts nothing. This trade-off has a clear biological reading: nnU-Net's errors are concentrated in the direction of under-counting, while U-Net's are split between under-counting and false detection. For analyses that treat empty regions as confirmed negatives, nnU-Net's behaviour is preferable. For analyses where missed detections are more costly than false positives, nnU-Net's conservatism becomes an issue. Zero-Dice cases in both models occur under the same conditions, namely sparse foreground or ambiguous local contrast, where predictions either miss entirely or fragment without reaching sufficient overlap. The low-density foreground quartile in the stratified analysis confirms this further.

5.4 Model Comparison

The direct comparison shows that nnU-Net outperforms the U-Net on the standard pixel-level metrics, and that the gap becomes larger once object-level behaviour is considered. nnU-Net leads in Dice, IoU, BF1, and finite HD95, while ECE differs only slightly. Table 5.9 summarises the key mean metrics at threshold 0.5 and shows how the object-level analysis sharpens that ranking.

Table 5.9: Key mean metrics at threshold 0.5 on the test set. Difference is reported as nnU-Net minus U-Net.

Metric	**U-Net**	**nnU-Net**	**nn − U**
Foreground Dice	0.578	0.624	+0.047
IoU	0.485	0.539	+0.053
Precision	0.671	0.729	+0.058
Recall	0.558	0.593	+0.035
BF1 (2 px)	0.626	0.684	+0.058
HD95 (paired finite mean, px)	43.06	39.78	−3.28
ECE	0.020	0.018	−0.002
Count error signed	0.04	−1.44	−1.48
Merge rate	0.041	0.035	−0.006
Split rate	0.091	0.041	−0.050
Connectivity error	0.132	0.076	−0.056
PQ (IoU $>$ 0.5)	0.459	0.544	+0.084
Obj. Precision @0.5	0.557	0.676	+0.119
Obj. Recall @0.5	0.558	0.606	+0.047

The pixel-level difference is modest but consistent. The object-level difference is larger, with nnU-Net leading by 0.084 in PQ and by 0.119 in object precision at IoU 0.5. Topology metrics follow the same pattern. U-Net has both a higher merge rate (0.041

vs. 0.035) and a much higher split rate (0.091 vs. 0.041), as well as a larger connectivity error (0.132 vs. 0.076).

The same pattern is also visible at the distribution level, with the U-Net Dice distribution shifted toward lower overlap (Appendix Figure A.2).

5.4.1 Trade-Offs

U-Net misses fewer GT-positive slices (278 vs. 331) and its signed count error is closer to zero overall, which might suggest comparable object recovery. However, that near-zero signed error is partly offset by fragmentation. U-Net has a much higher split rate (0.091 vs. 0.041) and connectivity error (0.132 vs. 0.076). The earlier pixel summaries also show more zero-Dice failures (551 vs. 493) and more false positives on GT-empty slices (160 vs. 78). Fewer complete misses therefore do not translate into better overall object recovery.

Across the 27 paired metrics tested in the evaluation, two-sided Wilcoxon signed-rank tests with Benjamini–Hochberg correction favour nnU-Net in all cases. This includes pixel overlap, boundary quality, calibration, and object-level metrics. These results provide strong paired evidence within the test set. However, because the tests are performed at slice level rather than biological-source level, they support the observed comparison but do not provide formal source-level inference.

5.4.2 Practical Importance

The Dice gap of 0.047 is relatively small when considered on its own. The object-level gaps are larger: PQ differs by 0.084, object precision at IoU 0.5 by 0.119, and split rate by more than a factor of two (0.091 vs. 0.041). These differences reflect U-Net's stronger tendency to fragment individual mitochondria into multiple predicted components. That would make object counts, morphology summaries, and the separation of adjacent mitochondria less reliable.

The advantage is also stable across threshold points. As shown in Figure 5.1, U-Net underperforms nnU-Net across the full threshold range from 0.30 to 0.70, so the ranking does not depend on threshold 0.5 alone.

A supplementary Dice–PQ scatter is included in Figure A.3. It shows the same pattern from another angle: Dice and PQ are positively related for both models, and the spread remains wide, especially in the high-Dice region.

5.4.3 High-Dice Corner Cases

This subsection addresses the third research focus: whether a slice that appears strong under pixel-level evaluation can still fail at the level of individual mitochondria. The corner-case diagnostic uses Dice > 0.85 as a high-overlap threshold and then asks whether the same slice shows object count error > 4, merge rate > 0, or split rate > 0. Across all slices, 373 of 2,854 U-Net cases (13.1%) and 436 of 2,772 nnU-Net cases (15.7%) meet this rule. Restricting the view to the high-Dice (> 0.85) subset shows why

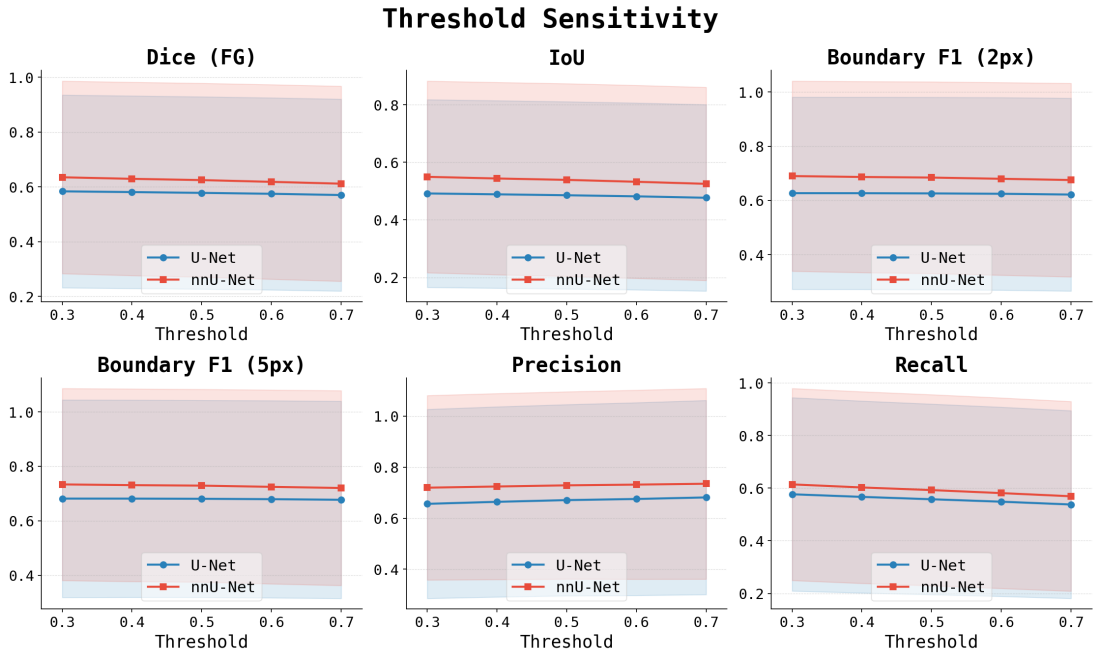


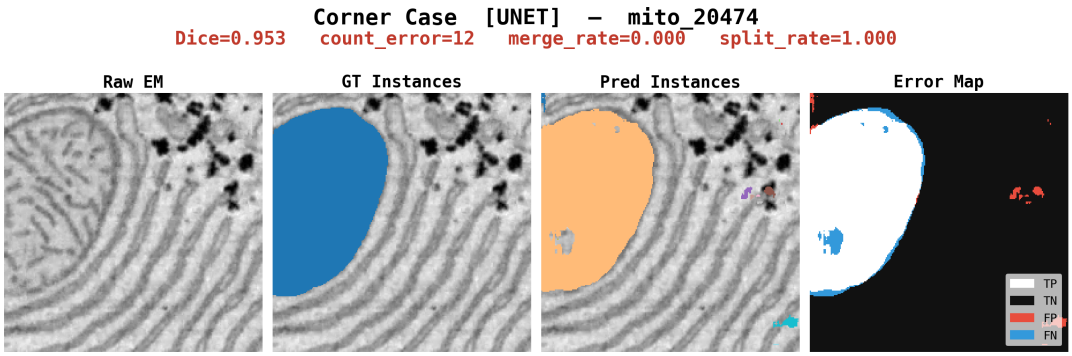
Figure 5.1: Threshold sensitivity curves for the test set. The ranking between U-Net and nnU-Net remains stable across the operating range 0.30–0.70.

Dice alone is not a safe acceptance criterion. 373 of 882 U-Net high-Dice slices (42.3%) and 436 of 1,144 nnU-Net high-Dice slices (38.1%) still show elevated object-level error.

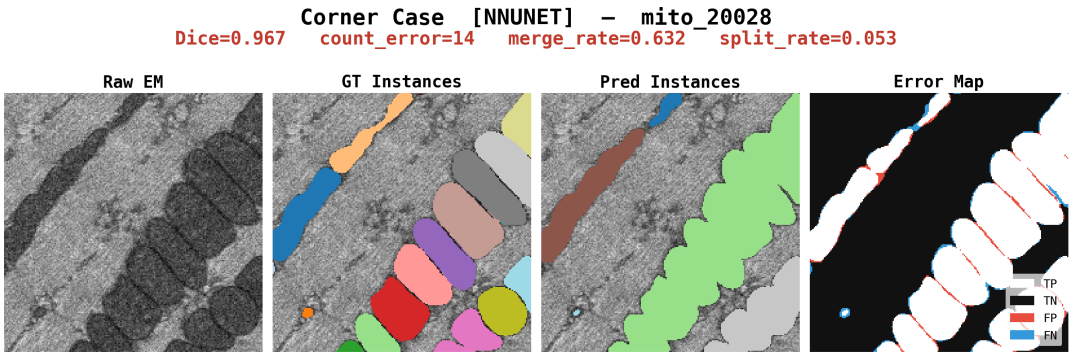
If Dice > 0.85 were used as a quality threshold, about two in five accepted slices would still contain a biologically meaningful error. These errors include count errors greater than four, merged clusters, fragmented objects, or combinations of these. Any biological analysis that depends on object count, morphology, or spatial separation would then be affected by those errors.

Figure 5.2 shows representative high-Dice failure modes directly. In the U-Net example, the dense cluster is over-fragmented despite Dice 0.953, with count error 12 and split rate 1.000. In the nnU-Net example, most of the cluster is merged despite Dice 0.967, producing count error 14 and merge rate 0.632. These are structural errors. One case is driven by fragmentation and the other by merging. The diagnostic uses Dice as the entry criterion, but the broader conclusion concerns pixel-level sufficiency: strong pixel-level scores do not guarantee reliable object recovery.

Figure 5.3 shows two further examples where the main difference lies in object separation, while overall coverage remains similar. In example 1, U-Net merges neighbouring mitochondria that nnU-Net keeps separate, despite similar Dice, producing a PQ gap of 0.608 vs. 0.910. In example 2, U-Net draws coarser boundaries in the denser region, reducing both Dice and PQ (0.648/0.341 vs. 0.746/0.527). These examples make visible the mechanism behind the aggregate PQ gap in Table 5.9: nnU-Net more reliably separates adjacent mitochondria as distinct objects.

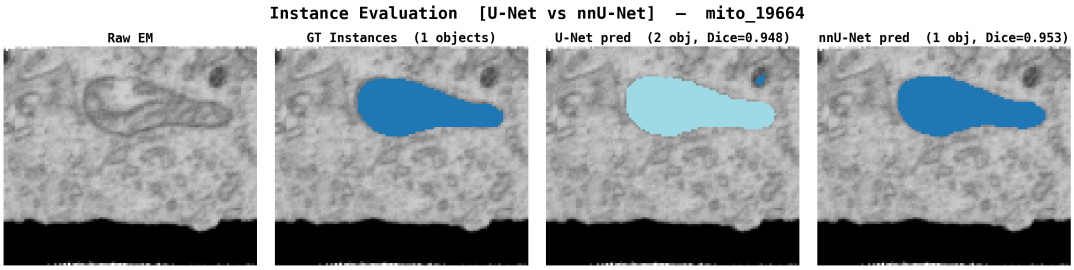


(a) U-Net corner case: Dice 0.953, count error 12, split rate 1.000.

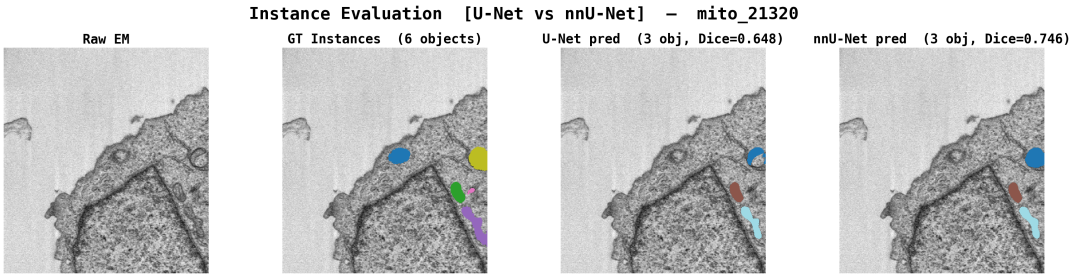


(b) nnU-Net corner case: Dice 0.967, count error 14, merge rate 0.632.

Figure 5.2: Representative high-Dice corner cases from the test set. Both predictions score above 0.95 on Dice, yet both contain severe topology errors.



(a) Qualitative example 1: similar Dice; PQ 0.608 vs. 0.910.



(b) Qualitative example 2: denser slice; Dice/PQ 0.648/0.341 vs. 0.746/0.527.

Figure 5.3: Representative qualitative comparisons from the test set. In both examples nnU-Net preserves the separation of adjacent mitochondria more clearly.

5.5 Stratified Analysis

Table 5.10 stratifies GT-positive test images by foreground pixel ratio quartile, with 674, 673, 673, and 674 images in Q1–Q4 respectively. Foreground ratio is the fraction of image pixels labelled as mitochondria in the ground truth. The quartile boundaries are derived from the GT-positive test set: Q1 (≤ 0.015), Q2 (0.015–0.040), Q3 (0.040–0.102), and Q4 (> 0.102).

Table 5.10: Metrics stratified by foreground pixel-ratio quartile (GT-positive images only; $n \approx 674$ per quartile). The object-count-error, split-rate, and PQ columns use the connected-component analysis. 'U' denotes U-Net and 'nn' denotes nnU-Net.

Quartile	U Dice	nn Dice	U Count	nn Count	U Split	nn Split	U PQ	nn PQ
Q1 (≤ 0.015)	0.398	0.435	2.202	1.426	0.039	0.020	0.290	0.325
Q2 (0.015–0.040)	0.570	0.605	2.905	2.574	0.067	0.051	0.385	0.445
Q3 (0.040–0.102)	0.722	0.767	3.798	2.730	0.109	0.046	0.486	0.591
Q4 (>0.102)	0.759	0.763	5.252	5.545	0.231	0.083	0.419	0.510

Dice FG improves strongly with foreground density for both models, from 0.398 to 0.759 for U-Net and from 0.435 to 0.763 for nnU-Net. In the sparse first quartile, the objects are small enough that even a few misplaced pixels can reduce Dice. Performance then improves through Q2 and Q3 as the objects become easier to localise and overlap.

Object-level quality follows a different pattern. PQ peaks in Q3 for both models (0.486 for U-Net and 0.591 for nnU-Net) before falling in Q4, while split rate rises sharply in the densest quartile, especially for U-Net (0.231 vs. 0.083). PQ falls in Q4 despite stronger Dice. Dense packing increases merges and splits, which PQ reflects directly but Dice is less sensitive to. At the pixel level, the Dice gap in Q4 is small (0.759 vs. 0.763), which would make the models appear nearly equivalent if only overlap were considered. At the object level, the models remain clearly separated: PQ stays higher for nnU-Net (0.510 vs. 0.419) and the split-rate gap widens substantially in Q4. Object count error also rises with density for both models, reaching 5.252 for U-Net and 5.545 for nnU-Net in the densest quartile.

Appendix Figure A.4 shows this pattern. The Dice curves draw closer in Q4, whereas split rate and PQ continue to separate the models. A smaller pixel-level gap in dense images does not imply that the two models perform equally well once individual mitochondria are evaluated at the object level. The dense-slice comparison shows the main result of the chapter most clearly: pixel-level differences do not capture the object-level difference between the models.

Chapter 6

Conclusions

The central methodological contribution of this dissertation is a unified, reproducible evaluation framework for EM mitochondrial segmentation. It is composed of four parts: a standardised comparison pipeline, a biologically-motivated metric set, a corner-case diagnostic, and a stratified analysis of performance across image conditions.

The comparison confirms that nnU-Net is the stronger model, but the more informative result concerns where the difference is concentrated. The foreground overlap gap is modest and consistent across thresholds. The object-level gap is larger, and it is clearest in dense images, where adjacent mitochondria are harder to separate. Pixel-overlap metrics make the models appear more similar, while panoptic quality and split rate distinguish the models. Overlap metrics alone do not show this pattern.

Beyond the overall ranking, the object-level analysis reveals that the two models fail in structurally distinct ways. U-Net's characteristic error is fragmentation. It over-segments single mitochondria into multiple predicted components, inflating object counts and distorting morphology statistics. nnU-Net is more prone to under-counting, systematically missing objects rather than splitting them. These are opposite errors with opposite biological consequences, yet both are consistent with moderate or strong foreground Dice. Overlap metrics do not distinguish between them.

The corner-case diagnostic confirms that strong overlap scores are not a reliable indicator of structural correctness. Among high-overlap predictions, a substantial proportion from both models still contained failures including merges, splits, or large count discrepancies. The type of failure depended on the model: U-Net failures in this range are predominantly splits, while nnU-Net failures more often involve merging. Predictions that appear comparable under an overlap threshold can carry qualitatively different biological errors, a distinction that is not clear from pixel-level summaries alone.

In practice, this means an EM segmentation study in which individual mitochondria need to be counted and separated should go beyond pixel-level summaries. Dice, IoU, BF1, HD95, and ECE remain useful. However, they should be reported alongside boundary-aware and object-level measures such as PQ, count error, and merge or split statistics whenever biological interpretation depends on individual mitochondria and their separation.

6.1 Limitations

Model formulation. Both models predict foreground as a binary mask, and individual mitochondria are recovered afterward by connected-component post-processing. The separation step is therefore not learned directly during training. The reported object-level results should not be taken as the best instance-level performance that more specialised instance segmentation models could achieve.

Scope and generalisability. The pipeline presented in this dissertation should be understood as a proof of concept for a broader evaluation strategy. The empirical study is bounded to a single dataset, so the findings should not be assumed to generalise beyond the image conditions, annotations, and biological variation represented in CEM-MitoLab. The central methodological point of the dissertation should be tested on a wider range of architectures and model families, whether semantic or instance-aware. That would help distinguish conclusions that are robust across modelling choices from those that are specific to the models examined. Formal paired significance testing is therefore informative within this dataset, but should not be overgeneralised beyond this experimental setting.

6.2 Future Work

1. Compare the current semantic models against instance-aware alternatives, to test whether end-to-end instance modelling reduces the merge, split, and count errors highlighted in this dissertation.
2. Apply the evaluation framework to additional EM datasets and model families, to determine which findings are specific to CEM-MitoLab and which transfer across imaging conditions and biological sources.
3. Extend the pipeline and analysis to 3D volumetric settings, where object-level errors along the z-axis can become more consequential than in slice-wise 2D evaluation.

Bibliography

Anusha Aswath, Ahmad Alsahaf, Ben N. G. Giepmans, and George Azzopardi. Segmentation in large-scale cellular electron microscopy with deep learning: A literature survey. *Medical Image Analysis*, 89:102920, 2023. doi: 10.1016/j.media.2023.102920.

Yoav Benjamini and Yosef Hochberg. Controlling the false discovery rate: A practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society: Series B (Methodological)*, 57(1):289–300, 1995. doi: 10.1111/j.2517-6161.1995.tb02031.x.

Vincent Casser, Kenny Kang, Hanspeter Pfister, and Daniel Haehn. Fast mitochondria detection for connectomics. In *Proceedings of the 1st Conference on Medical Imaging with Deep Learning*, volume 121 of *Proceedings of Machine Learning Research*, pages 111–120. PMLR, 2020. URL https://proceedings.mlr.press/v121/casser20a.html.

Chaoyi Chen, Yidan Yan, Jingpeng Wu, and Wen-Biao Gan. GCTransNet: 3D mitochondrial instance segmentation based on global context vision transformers. *Journal of Structural Biology*, 217:108170, 2025. doi: 10.1016/j.jsb.2025.108170.

Bowen Cheng, Maxwell D. Collins, Yukun Zhu, Ting Liu, Thomas S. Huang, Hartwig Adam, and Liang-Chieh Chen. Panoptic-DeepLab: A simple, strong, and fast baseline for bottom-up panoptic segmentation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 12475–12485, 2020. doi: 10.1109/CVPR42600.2020.01249. URL https://openaccess.thecvf.com/content_CVPR_2020/html/Cheng_Panoptic-DeepLab_A_Simple_Strong_and_Fast_Baseline_for_Bottom-Up_Panoptic_CVPR_2020_paper.html.

Ryan Conrad and Kedar Narayan. Instance segmentation of mitochondria in electron microscopy images with a generalist deep learning model trained on a diverse dataset. *Cell Systems*, 14:58–71.e5, 2023. doi: 10.1016/j.cels.2022.12.006.

Daniel Franco-Barranco, Arrate Muñoz-Barrutia, and Ignacio Arganda-Carreras. Stable deep neural network architectures for mitochondria segmentation on electron microscopy volumes. *Neuroinformatics*, 20:437–450, 2022. doi: 10.1007/s12021-021-09556-1.

Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger. On calibration of modern neural networks. In *Proceedings of the 34th International Conference on Machine*

Learning, volume 70 of *Proceedings of Machine Learning Research*, pages 1321–1330. PMLR, 2017. URL https://proceedings.mlr.press/v70/guo17a.html.

Larissa Heinrich, Davis Bennett, David Ackerman, Woohyun Park, John Bogovic, Nils Eckstein, Alyson Petruncio, Jody Clements, Song Pang, C. Shan Xu, Jan Funke, Wyatt Korff, Harald F. Hess, Jennifer Lippincott-Schwartz, Stephan Saalfeld, Aubrey V. Weigel, and COSEM Project Team. Whole-cell organelle segmentation in volume electron microscopy. *Nature*, 599:141–146, 2021. doi: 10.1038/s41586-021-03977-3.

Yuyao Huang, Nickhil Jadav, Georgia Rutter, Lech Szymanski, Mihnea Bostina, and Duane P. Harland. A generalist deep-learning volume segmentation tool for volume electron microscopy of biological samples. *Journal of Structural Biology*, 217: 108214, 2025. doi: 10.1016/j.jsb.2025.108214.

Fabian Isensee, Paul F. Jaeger, Simon A. A. Kohl, Jens Petersen, and Klaus H. Maier-Hein. nnU-Net: a self-configuring method for deep learning-based biomedical image segmentation. *Nature Methods*, 18:203–211, 2021. doi: 10.1038/s41592-020-01008-z.

Chan Jang, Hojun Lee, Jaejun Yoo, and Haejin Yoon. Deep learning-driven automated mitochondrial segmentation for analysis of complex transmission electron microscopy images. *Scientific Reports*, 15:19076, 2025. doi: 10.1038/s41598-025-03311-1.

Alexander Kirillov, Kaiming He, Ross Girshick, Carsten Rother, and Piotr Dollár. Panoptic segmentation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 9396–9405, 2019. doi: 10.1109/CVPR.2019.00963.

Harold W. Kuhn. The hungarian method for the assignment problem. *Naval Research Logistics Quarterly*, 2(1–2):83–97, 1955. doi: 10.1002/nav.3800020109.

Ilya Loshchilov and Frank Hutter. Decoupled weight decay regularization. In *International Conference on Learning Representations*, 2019. URL https://openreview.net/forum?id=Bkg6RiCqY7.

Aurélien Lucchi, Kevin Smith, Radhakrishna Achanta, Graham Knott, and Pascal Fua. Supervoxel-based segmentation of mitochondria in em image stacks with learned shape features. *IEEE Transactions on Medical Imaging*, 31(2):474–486, 2012. doi: 10.1109/TMI.2011.2171705.

Andrew L. Maas, Awni Y. Hannun, and Andrew Y. Ng. Rectifier nonlinearities improve neural network acoustic models. In *Proceedings of the 30th International Conference on Machine Learning, Workshop on Deep Learning for Audio, Speech and Language Processing*, 2013. URL https://ai.stanford.edu/~amaas/papers/relu_hybrid_icml2013_final.pdf.

David Martin, Charless Fowlkes, Doron Tal, and Jitendra Malik. A database of human segmented natural images and its application to evaluating segmentation algorithms and measuring ecological statistics. In *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, pages II: 416–423, 2001. URL https://vision.ics.uci.edu/papers/MartinFTM_ICCV_2001/.

Fausto Milletari, Nassir Navab, and Seyed-Ahmad Ahmadi. V-net: Fully convolutional neural networks for volumetric medical image segmentation. In *2016 Fourth International Conference on 3D Vision (3DV)*, pages 565–571, 2016. doi: 10.1109/3DV.2016.79.

James Munkres. Algorithms for the assignment and transportation problems. *Journal of the Society for Industrial and Applied Mathematics*, 5(1):32–38, 1957. doi: 10.1137/0105003.

Vinod Nair and Geoffrey E. Hinton. Rectified linear units improve restricted boltzmann machines. In *Proceedings of the 27th International Conference on Machine Learning (ICML)*, pages 807–814, 2010.

Kedar Narayan. EMPIAR-11037: CEM-MitoLab: a dataset of 22k cellular EM 2d images with label maps of 135k mitochondrial instances, for deep learning. Electron Microscopy Public Image Archive (EMPIAR), 2022. URL `https://empiar.pdbj.org/en/entry/11037`.

Olaf Ronneberger, Philipp Fischer, and Thomas Brox. U-Net: Convolutional networks for biomedical image segmentation. In *Medical Image Computing and Computer-Assisted Intervention – MICCAI 2015*, volume 9351 of *Lecture Notes in Computer Science*, pages 234–241. Springer, Cham, 2015. doi: 10.1007/978-3-319-24574-4_28.

Anu Suomalainen and Jodi Nunnari. Mitochondria at the crossroads of health and disease. *Cell*, 187, may 2024. doi: 10.1016/j.cell.2024.04.037. Leading Edge Review; Open Access.

Abdel Aziz Taha and Allan Hanbury. Metrics for evaluating 3D medical image segmentation: analysis, selection, and tool. *BMC Medical Imaging*, 15:29, 2015. doi: 10.1186/s12880-015-0068-x.

Lisa Tilokani, Shun Nagashima, Vincent Paupe, and Julien Prudent. Mitochondrial dynamics: overview of molecular mechanisms. *Essays in Biochemistry*, 62:341–360, 2018. doi: 10.1042/EBC20170104.

Donglai Wei, Zudi Lin, Daniel Franco-Barranco, Nils Wendt, Xingyu Liu, Wenjie Yin, Xin Huang, Aarush Gupta, Won-Dong Jang, Xueying Wang, Ignacio Arganda-Carreras, Jeff W. Lichtman, and Hanspeter Pfister. MitoEM dataset: Large-scale 3D mitochondria instance segmentation from EM images. In *Medical Image Computing and Computer-Assisted Intervention – MICCAI 2020*, volume 12265 of *Lecture Notes in Computer Science*, pages 66–76. Springer, Cham, 2020. doi: 10.1007/978-3-030-59722-1_7.

Frank Wilcoxon. Individual comparisons by ranking methods. *Biometrics Bulletin*, 1 (6):80–83, 1945. doi: 10.2307/3001968.

Ronald Xie, Ben Mulcahy, Ali Darbandi, Sagar Marwah, Fez Ali, Yuna Lee, Gunes Parlakgul, Gokhan S. Hotamisligil, Bo Wang, Sonya MacParland, Mei Zhen, and Gary D. Bader. Transfer learning improves performance in volumetric electron

microscopy organelle segmentation across tissues. *Bioinformatics Advances*, 5(1): vbaf021, 2025. doi: 10.1093/bioadv/vbaf021.

Özgün Çiçek, Ahmed Abdulkadir, Soeren S. Lienkamp, Thomas Brox, and Olaf Ronneberger. 3D U-Net: Learning dense volumetric segmentation from sparse annotation. In *Medical Image Computing and Computer-Assisted Intervention – MICCAI 2016*, volume 9901 of *Lecture Notes in Computer Science*, pages 424–432. Springer, Cham, 2016. doi: 10.1007/978-3-319-46723-8_49.

Appendix A

Supplementary Results

This appendix collects supplementary statistical outputs, extended result tables, and additional qualitative examples that support the main analysis in Chapter 5.

A.1 Count Error Distribution

Figure A.1 shows the distribution of absolute object count error over GT-positive test slices for both models, with the corner-case cutoff of greater than 4 marked. The majority of slices fall below the cutoff, and the distributions are right-skewed with a long tail of large errors. The dashed line sits at the 75th percentile for nnU-Net (Q3 = 4) and just below the 75th percentile for U-Net (Q3 = 5), confirming that the threshold targets the upper tail rather than typical per-slice variation.

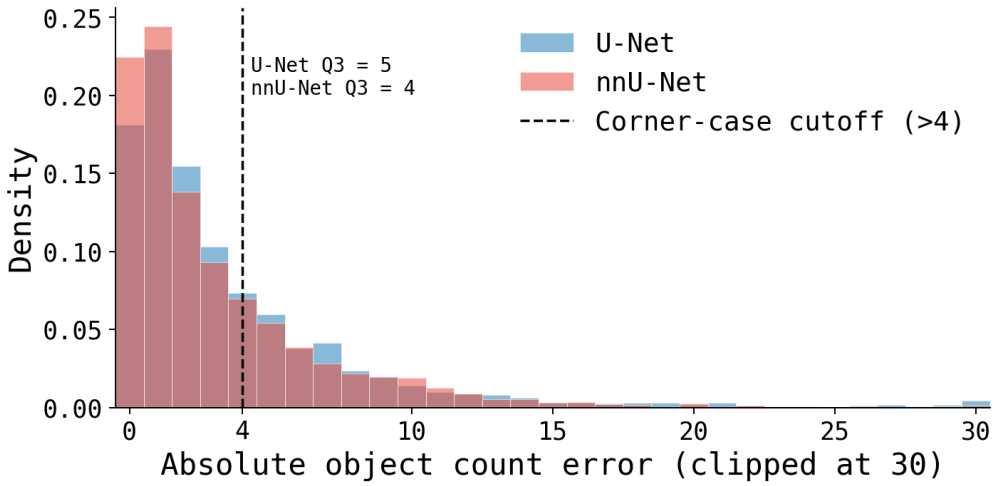


Figure A.1: Absolute object count error distributions over GT-positive test slices (values clipped at 30 for display). The dashed line marks the corner-case cutoff of greater than 4. Both distributions are right-skewed; the cutoff sits at or just above the 75th percentile for each model.

A.2 Full Paired Significance Tests

Tables A.1 and A.2 report the full Wilcoxon signed-rank test results with Benjamini–Hochberg-adjusted p-values for the paired evaluation. The difference column is reported as nnU-Net minus U-Net, so positive values favour nnU-Net for higher-is-better metrics and negative values favour nnU-Net for lower-is-better metrics.

Table A.1: Supplementary paired significance tests for pixel-level, boundary, and calibration metrics.

Metric	**U-Net**	**nnU-Net**	**nn − U**	**BH-adjusted p**
Dice FG @0.5	0.599	0.629	+0.030	1.04×10^{-38}
Dice BG @0.5	0.985	0.987	+0.002	3.06×10^{-87}
IoU @0.5	0.503	0.543	+0.039	4.45×10^{-46}
Precision @0.5	0.696	0.734	+0.038	1.71×10^{-75}
Recall @0.5	0.578	0.597	+0.019	7.99×10^{-13}
HD95 @0.5	43.060	39.783	-3.277	1.97×10^{-11}
BF1 2px @0.5	0.626	0.684	+0.058	1.02×10^{-62}
BF1 5px @0.5	0.681	0.729	+0.048	1.07×10^{-39}
Dice FG @0.3	0.606	0.641	+0.035	6.30×10^{-48}
Dice FG @0.4	0.603	0.635	+0.032	2.43×10^{-43}
Dice FG @0.6	0.595	0.622	+0.027	9.42×10^{-34}
Dice FG @0.7	0.590	0.615	+0.025	1.58×10^{-29}
ECE	0.020	0.018	-0.002	2.22×10^{-46}

Table A.2: Supplementary paired significance tests for object-level metrics.

Metric	**U-Net**	**nnU-Net**	**nn − U**	**BH-adjusted p**
Abs count error	3.066	2.559	-0.507	1.73×10^{-12}
Merge rate	0.041	0.035	-0.006	3.15×10^{-8}
Split rate	0.091	0.041	-0.050	5.75×10^{-51}
Connectivity error	0.132	0.076	-0.056	4.25×10^{-57}
PQ @IoU 0.5	0.459	0.544	+0.084	2.03×10^{-116}
Mean matched IoU	0.482	0.514	+0.032	5.52×10^{-60}
Obj. precision @0.3	0.594	0.709	+0.115	3.36×10^{-106}
Obj. recall @0.3	0.593	0.633	+0.039	2.53×10^{-19}
Obj. precision @0.5	0.557	0.676	+0.119	1.50×10^{-109}
Obj. recall @0.5	0.558	0.606	+0.047	1.77×10^{-30}
Obj. precision @0.7	0.492	0.611	+0.119	1.92×10^{-114}
Obj. recall @0.7	0.494	0.556	+0.061	1.15×10^{-51}
Obj. precision @0.9	0.247	0.329	+0.082	3.43×10^{-95}
Obj. recall @0.9	0.246	0.308	+0.062	4.03×10^{-68}

A.3 Extended Result Tables

A.3.1 GT-Empty Slice Behaviour

Table A.3 isolates the 618 background-only test slices. This makes the difference in empty-slice behaviour explicit: U-Net produces false positives on 160 of 618 GT-empty slices (25.9%), whereas nnU-Net does so on 78 of 618 (12.6%). In the evaluator, foreground-only overlap metrics are undefined on empty-empty slices. Dice FG, IoU, precision, and recall are therefore 0 in this table because the only finite GT-empty cases are false-positive predictions. Boundary F1 remains nonzero because empty-empty slices are still treated as a perfect boundary match.

Table A.3: Supplementary results on GT-empty slices only at threshold 0.5.

Metric	**U-Net**	**nnU-Net**
Pred. foreground pixels (mean)	329.654	111.466
False-positive GT-empty slices	160 / 618	78 / 618
Dice FG (mean)	0.000	0.000
Dice BG (mean)	0.997	0.999
IoU (mean)	0.000	0.000
Precision (mean)	0.000	0.000
Recall (mean)	0.000	0.000
BF1 (2 px) mean	0.741	0.874
BF1 (5 px) mean	0.741	0.874
ECE (mean)	0.005	0.002

A.4 Supplementary Comparison Visuals

Figure A.2 gives the full foreground-Dice distribution comparison, while Figure A.3 shows the corresponding Dice–PQ relationship over individual test slices. Appendix Figure A.4 shows the foreground-density stratification discussed in Section 5.5. Appendix Figure A.5 provides the combined training convergence plot for the two models.

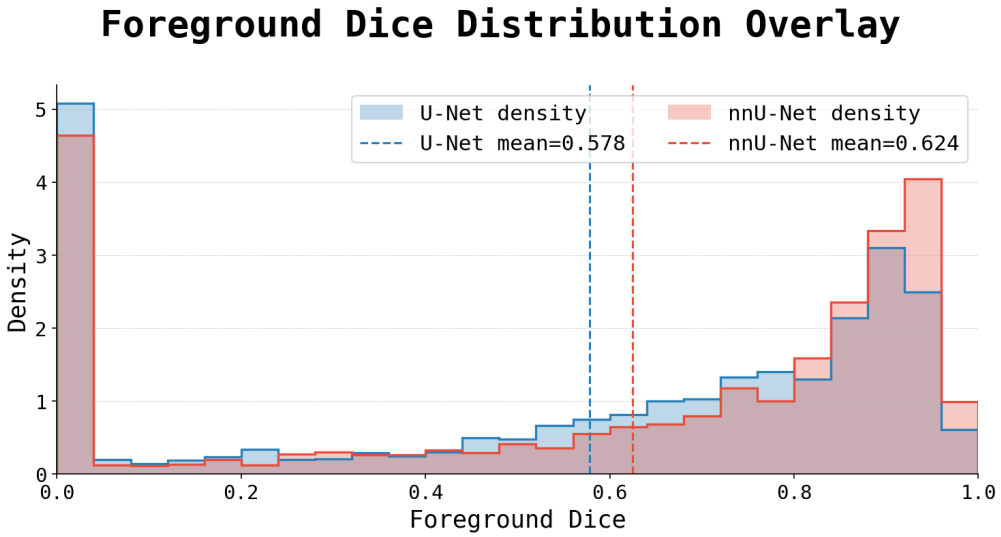


Figure A.2: Supplementary foreground Dice distributions for U-Net and nnU-Net over the test set. The U-Net distribution is shifted toward lower Dice, although both models retain a long left tail of difficult cases.

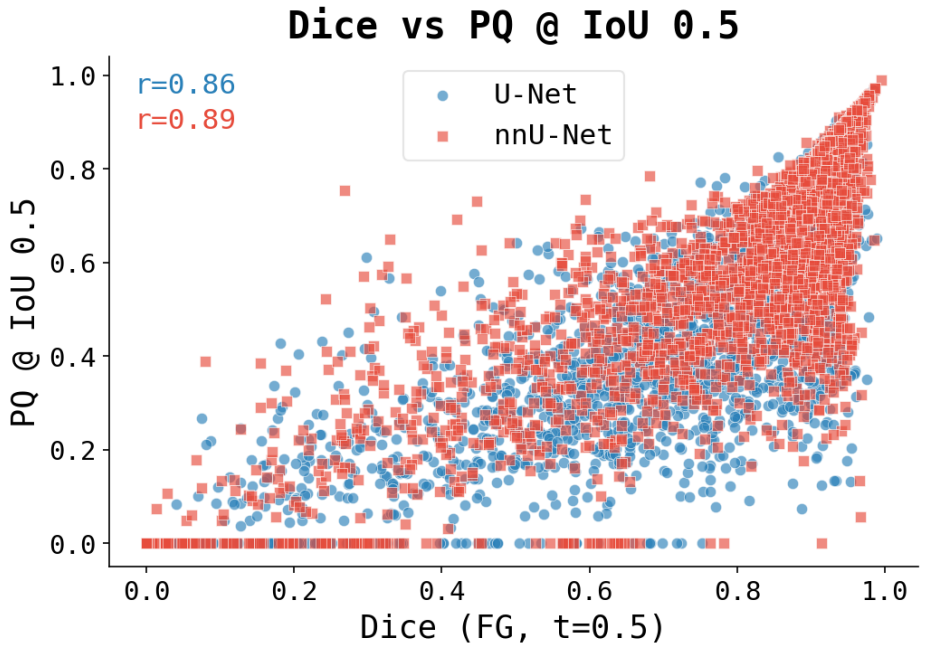


Figure A.3: Supplementary Dice–PQ scatter over the test set. Similar Dice values can still correspond to meaningfully different object-level quality.

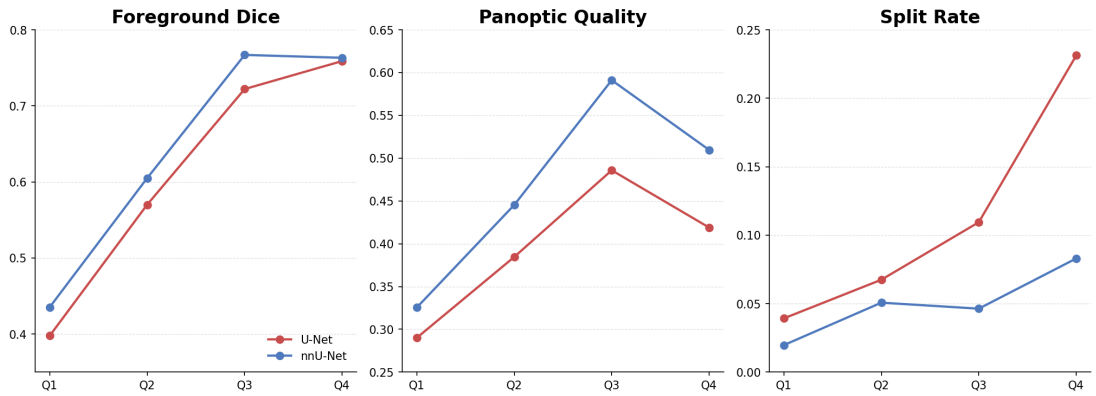


Figure A.4: Foreground-density stratification over GT-positive test slices. Dice improves with density for both models, but object-level quality peaks before the densest quartile and split rate rises sharply in Q4, especially for U-Net. The Dice curves converge in Q4 while the PQ and split-rate gaps widen.

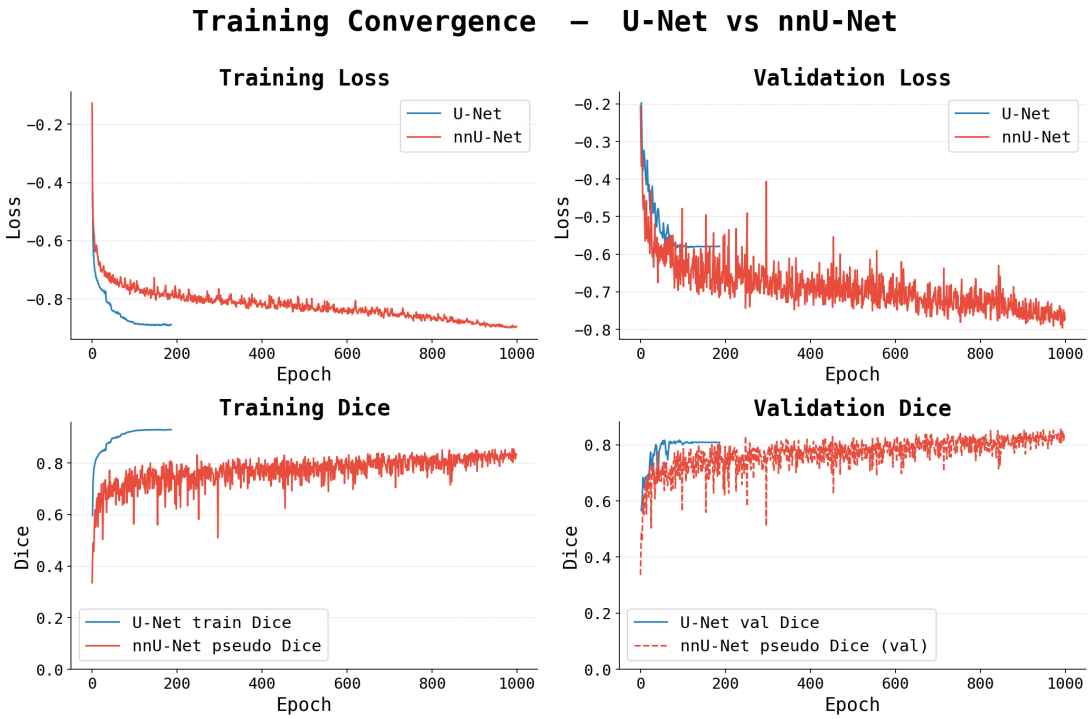
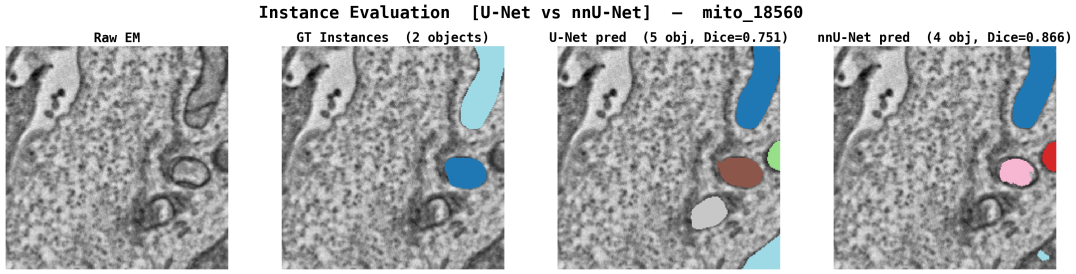


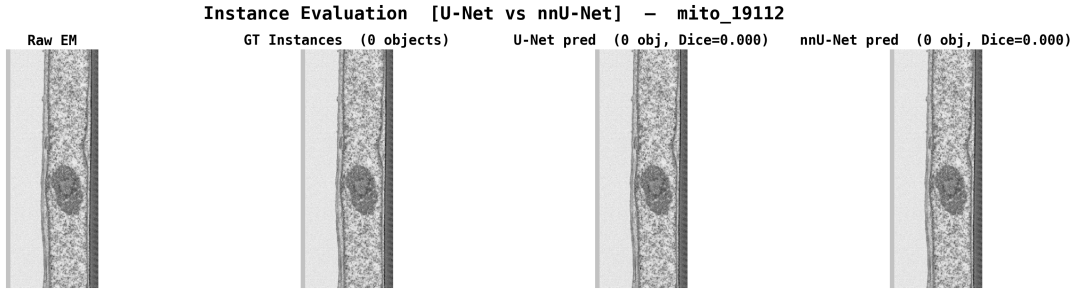
Figure A.5: Supplementary training convergence for the two experiments. The custom U-Net stops early under the validation-loss criterion, whereas nnU-Net completes the full 1,000-epoch schedule.

A.5 Additional Qualitative Examples

A.5.1 Additional Test Comparisons

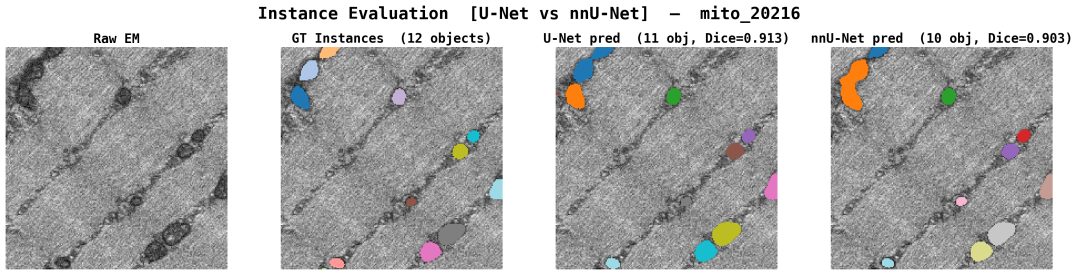


(a) `mito_18560`: additional test comparison example.

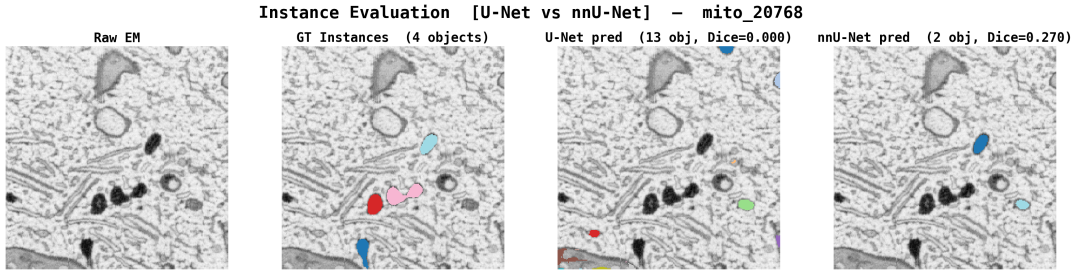


(b) `mito_19112`: additional test comparison example.

Figure A.6: Additional qualitative comparisons between U-Net and nnU-Net on the test set.



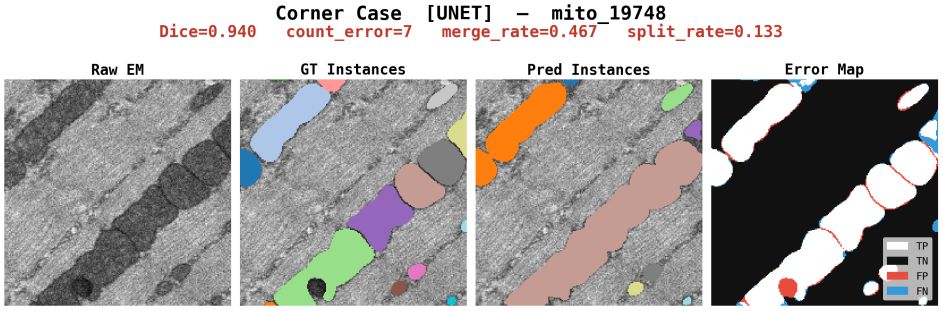
(a) `mito_20216`: additional test comparison example.



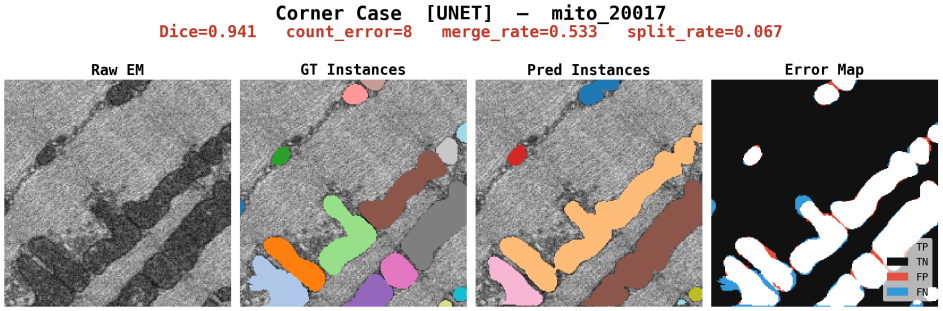
(b) `mito_20768`: additional test comparison example.

Figure A.7: Further qualitative comparisons between U-Net and nnU-Net on difficult test slices.

A.5.2 Additional Corner Cases

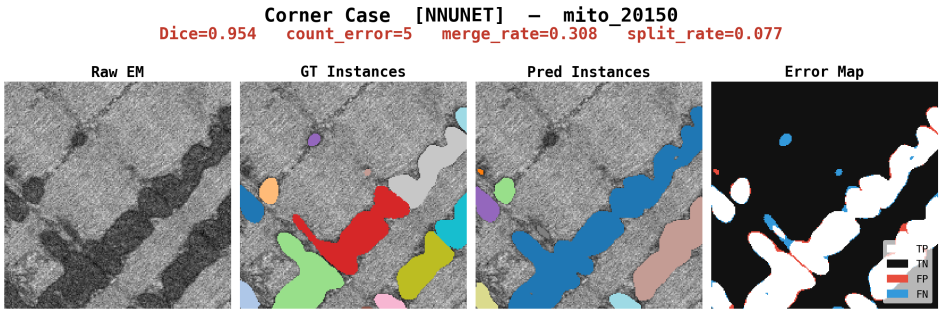


(a) `mito_19748`: additional U-Net corner case.

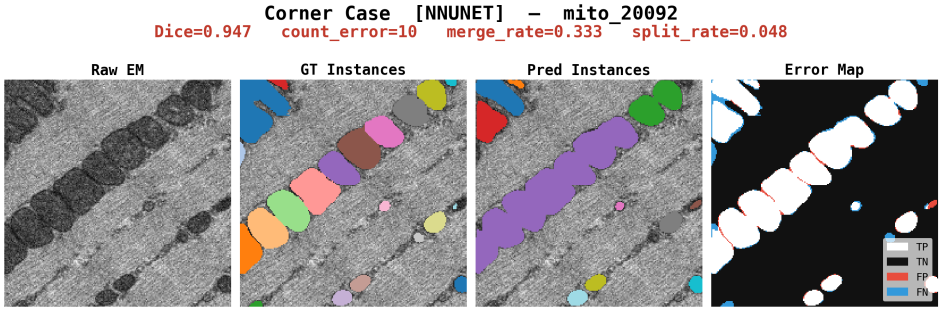


(b) `mito_20017`: additional U-Net corner case.

Figure A.8: Additional U-Net corner cases.



(a) `mito_20150`: additional nnU-Net corner case.



(b) `mito_20092`: additional nnU-Net corner case.

Figure A.9: Additional nnU-Net corner cases.